MBFT 814 : Microbial Biotechnology 3(2-0-2)

M.Sc. Microbiology, Semester III · Long-form detailed mid-term notes

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#	Topic (mid-term syllabus)	Main points covered
1	Introduction to microbial biotechnology	Definitions, why microbes, industrial strain, fermentation process components, media, products, scope
2	History of microbial biotechnology	Eras, detailed timeline, key discoveries explained, Indian milestones
3	Types of bioreactors	Parts with labelled diagram, ideal bioreactor, CSTR, bubble column, airlift, fluidized bed, packed bed, membrane, photobioreactor
4	Types of fermentation methods	Batch (growth curve, kinetics), fed-batch, continuous (chemostat, turbidostat), SmF, surface, SSF, semi-solid
5	Enzyme production	Types, regulation, steps, amylase and protease examples, downstream flow chart, units, applications
6	Immobilization of enzymes	History, supports, 5 methods with procedure and diagram, properties, reactors, applications
7	Screening and strain improvement	Primary and secondary screening, mutagens and their action, mutant selection, protoplast fusion, rDNA
8	Antibiotic production	Classification; penicillin (biosynthesis, process, flow chart); streptomycin; tetracycline; griseofulvin
9	Vitamin production	Riboflavin (B_2), cyanocobalamin (B_{12}), vitamin C, β-carotene
10	Quick revision and likely questions	One-line facts and question bank

1. INTRODUCTION TO MICROBIAL BIOTECHNOLOGY

Detailed explanation

Microbial biotechnology is the part of biotechnology in which microorganisms are the working "factories". A microbial cell is a very small but complete chemical plant: it takes in simple, cheap nutrients such as sugar, ammonia and mineral salts, and with the help of hundreds of enzymes converts them into cell material and into products that are useful to us. When we grow such cells in very large numbers under controlled conditions, the small amount of product made by each cell adds up to tonnes. This is the basic idea behind every industrial fermentation, from beer and bread to antibiotics, vitamins and enzymes.

The subject brings together three disciplines. **Microbiology** tells us which organism to use and how it grows; **biochemistry** explains the pathways by which the product is formed and how these pathways are regulated; and **engineering** provides the equipment (fermenters, sterilizers, separators) needed to carry out the process on a large scale. This is why the European Federation of Biotechnology defines biotechnology as the *integrated* use of these sciences. A good microbial biotechnologist must understand all three, because a strain that gives an excellent yield in a flask can fail completely in a 100,000-litre tank if oxygen supply, heat removal or sterility are not properly engineered.

Microorganisms are preferred for industrial production for several reasons. First, they grow extremely fast. A bacterium such as *E. coli* can divide every 20 minutes, and yeast every one to two hours, so a large amount of biomass and product can be obtained within a few days, whereas plants and animals need months or years. Second, because microbes are small, they have a very large surface area in relation to their volume. Nutrients enter and products leave very quickly, which gives them a very high rate of metabolism. Third, microorganisms are metabolically extremely diverse. As a group they can use almost any organic compound, and even carbon dioxide or methane, as food, and they can produce a huge range of compounds, including many that cannot be made economically by chemical synthesis (for example, complex antibiotics). Fourth, microbes can be grown on cheap raw materials and even on wastes such as molasses, whey, corn steep liquor and straw, which lowers the cost of the product. Finally, microbes are easy to manipulate genetically, so their yields can be raised many-fold by mutation and recombinant DNA technology, and production is not affected by climate or season.

Any industrial fermentation, whatever the product, has the same basic parts. A suitable medium is formulated and sterilized; a pure, active inoculum is developed through several stages; the organism is grown in the production fermenter under carefully controlled conditions; the product is recovered and purified (downstream processing); and finally the wastes are treated. The steps up to fermentation are called **upstream processing** and the recovery steps are called **downstream processing**. For many products, especially high-value ones like antibiotics and enzymes, downstream processing accounts for more than half of the total production cost.

Microbial products are of different kinds. Some processes aim at the **cells themselves** (baker's yeast, single cell protein, bacterial biofertilizers); others at **enzymes** (amylase, protease); others at **metabolites**. Metabolites are divided into primary and secondary metabolites. Primary metabolites are compounds that the cell needs for growth, such as amino acids, nucleotides, vitamins, organic acids and ethanol; they are made during the active growth phase, called the **trophophase**. Secondary metabolites, such as antibiotics, pigments and toxins, are not needed for growth; they are usually made after the growth rate falls, in the phase called the **idiophase**, and are produced only by certain organisms. This difference is very important in practice, because it decides how the fermentation is run: a primary metabolite is favoured by conditions that support fast growth, whereas for a secondary metabolite the organism is first grown and then shifted to production conditions, often by limiting a nutrient such as phosphate or by slow feeding of sugar.

1.1 Definitions

- **Biotechnology**: the word was first used by the Hungarian engineer **Karl Ereky in 1919** for the production of products from raw materials with the help of living organisms.
- **OECD / Convention on Biological Diversity (1992)**: "Any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use."

- **European Federation of Biotechnology**: "the integrated use of **biochemistry, microbiology and engineering sciences** in order to achieve technological (industrial) application of the capabilities of microorganisms, cultured tissue cells and parts thereof".
- **Microbial biotechnology**: the branch that uses **microorganisms** (bacteria, actinomycetes, fungi, yeasts, microalgae), their enzymes or their genes to make useful products (food, drugs, chemicals, fuels) or to give services (waste treatment, bioremediation) on a commercial scale.
- **Fermentation (industrial meaning)**: any process in which microorganisms are grown on a large scale to form a product, whether aerobic or anaerobic. (In the biochemical meaning, fermentation is anaerobic energy-yielding metabolism where organic compounds act as electron donor and acceptor.)

1.2 Why microorganisms are used

Property	Why it matters in industry
Small size, large surface : volume ratio	Very fast uptake of nutrients and release of products, so high metabolic rate
Rapid growth (generation time 20 min to a few hours)	Large biomass in hours or days instead of months (plants, animals)
Metabolic diversity	Can use sugars, starch, cellulose, hydrocarbons, methanol, CO_2; make acids, alcohols, enzymes, antibiotics, vitamins, polymers
Grow on cheap raw materials	Molasses, corn steep liquor, whey, bran, agro-industrial wastes lower the cost
Easy genetic manipulation	Mutation and recombinant DNA can raise yield many times
Controlled production	Independent of climate, season and land; uniform quality; easy scale-up in fermenters

1.3 Characteristics of an ideal industrial microorganism

- Pure culture, genetically **stable**, and easy to preserve for a long time.
- Gives a **high yield** of the required product in a **short time**.
- Grows on **cheap**, easily available medium; forms few unwanted by-products.
- Non-pathogenic, non-toxic, preferably **GRAS** (Generally Regarded As Safe).
- Produces spores or cells that are easy to use as **inoculum**.
- Product is easy to separate (preferably **extracellular**); resistant to contamination and phage.

1.4 Components (stages) of a fermentation process

Every industrial fermentation has six basic parts (Stanbury, Whitaker and Hall):

1. **Formulation of media** for the inoculum and production stages.
2. **Sterilization** of the medium, fermenter and air (medium: batch at 121°C, or continuous high-temperature short-time; air: filtration).
3. **Inoculum development**: stock culture → slant → shake flask → seed fermenter → production fermenter (usually 3–10% v/v inoculum).
4. **Production** in the fermenter under optimum pH, temperature, aeration and agitation.
5. **Downstream processing**: separation, extraction, purification and formulation of the product.
6. **Effluent treatment**: disposal of spent broth and biomass.

Steps 1–4 are called **upstream processing**; step 5 is **downstream processing**.

1.5 Fermentation media

Component	Examples	Role
Carbon source	Glucose, sucrose, **molasses**, starch, lactose, whey, vegetable oils, methanol	Energy and carbon skeleton
Nitrogen source	**Corn steep liquor (CSL)**, soybean meal, yeast extract, peptone, ammonium salts, urea	Proteins, nucleic acids

Component	Examples	Role
Minerals	P, K, Mg, S, trace Fe, Zn, Mn, Co, Cu	Enzyme cofactors, cell structure
Growth factors	Vitamins, amino acids (often supplied by CSL, yeast extract)	Needed by fastidious strains
Precursors	Phenylacetic acid (penicillin G), glycine (riboflavin), cobalt + DMB (vitamin B_{12})	Built directly into the product, raise yield
Inducers	Starch (amylase), oils (lipase), cellulose (cellulase)	Switch on enzyme genes
Buffers / antifoams	$CaCO_3$, phosphates / silicone oils, vegetable oils	Control pH / control foam

1.6 Types of microbial products

- **Microbial cells (biomass)**: baker's yeast, single cell protein (SCP), biofertilizers, probiotics, vaccines.
- **Microbial enzymes**: amylase, protease, lipase, cellulase, glucose isomerase.
- **Primary metabolites**: ethanol, citric acid, lactic acid, amino acids, nucleotides, vitamins.
- **Secondary metabolites**: antibiotics, pigments, toxins, alkaloids, gibberellins.
- **Recombinant products**: insulin, interferons, growth hormone, hepatitis B vaccine.
- **Biotransformation products**: steroid conversions (progesterone to 11α-hydroxyprogesterone by *Rhizopus*), vitamin C step by *Gluconobacter*.

Primary metabolites	Secondary metabolites
Made during active growth: **trophophase** (log phase)	Made after growth slows: **idiophase** (late log / stationary)
Essential for growth and reproduction	Not essential for growth; help survival (competition, defence)
Common to most organisms; produced in the same way	Restricted to a few species or strains
Production is growth-linked	Production is non-growth-linked; often repressed by glucose, phosphate
e.g. ethanol, citric acid, amino acids, nucleotides, vitamins	e.g. penicillin, streptomycin, tetracycline, pigments, mycotoxins

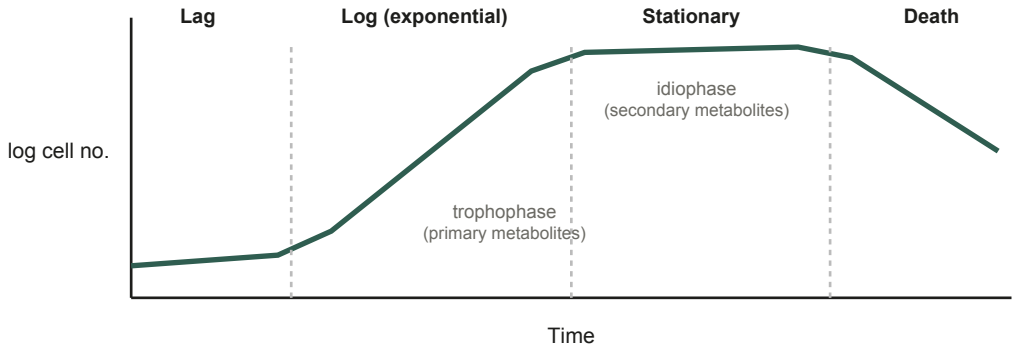


Fig. 1. Batch growth curve showing trophophase and idiophase

1.7 Scope and applications

Area	Examples
Food and beverages	Beer, wine, bread, cheese, curd, vinegar, soy sauce, SCP, amino acids (glutamic acid as MSG, lysine), flavours
Health and pharmaceuticals	Antibiotics, vitamins, vaccines, insulin, interferons, monoclonal antibodies, steroids, therapeutic enzymes
Industrial chemicals	Ethanol, acetone, butanol, glycerol, citric, lactic, gluconic and itaconic acids, biopolymers (PHB, xanthan)
Enzymes	Detergent, starch, textile, leather, paper, dairy, juice industries
Agriculture	Biofertilizers (*Rhizobium*, *Azotobacter*, PSB), biopesticides (*Bacillus thuringiensis*, *Trichoderma*), silage, plant growth promoters

Area	Examples
Environment	Sewage and effluent treatment, bioremediation of oil and pesticides, biogas, composting, bioleaching of copper and uranium
Energy	Bioethanol, biogas (methane), biodiesel, biohydrogen, microbial fuel cells

Exam point: For a long answer: definitions (Ereky 1919 + OECD) → why microbes → ideal industrial strain → six components of a fermentation process → primary vs secondary metabolites with growth curve → scope table.

2. HISTORY OF MICROBIAL BIOTECHNOLOGY

Detailed explanation

The history of microbial biotechnology can be understood as a gradual change from **using microbes without knowing about them** to **controlling them deliberately at the level of their genes**. For thousands of years people made beer, wine, bread, curd, cheese and vinegar. They did not know that microorganisms were responsible; they only knew from experience that certain practices gave good products. Sumerians and Babylonians brewed beer about 6000 BC, Egyptians made leavened bread about 4000 BC, and cheese making with rennet from calf stomach is several thousand years old. Fermentation was also used as a method of preserving food, because the acid or alcohol formed prevented spoilage.

The scientific basis began in the seventeenth century, when **Antonie van Leeuwenhoek** first saw microorganisms with his simple microscope (1673–1676). The real turning point, however, came with **Louis Pasteur** in the nineteenth century. Pasteur showed that fermentation is a biological process carried out by living microorganisms, that different microbes carry out different fermentations (yeast gives alcohol, certain bacteria give lactic acid), and that spoilage of wine and beer is caused by unwanted microbes. His method of gentle heating to kill spoilage organisms, pasteurization, is still used. At about the same time **Robert Koch** developed methods for obtaining pure cultures on solid media. Pure cultures are the foundation of industrial microbiology, because a process can only be reproducible if it is carried out with a single known organism.

In 1897 **Eduard Buchner** made a very important discovery: a juice extracted from crushed yeast cells, containing no living cells, could still ferment sugar to alcohol. This proved that fermentation is carried out by **enzymes**, and it opened the way for enzyme technology. Early in the twentieth century industrial fermentation developed rapidly, often because of the needs of war. During the First World War, **Chaim Weizmann** used *Clostridium acetobutylicum* to produce acetone and butanol from maize starch; acetone was needed to make the explosive cordite. This was the first large industrial process that needed strict aseptic (contamination-free) conditions. In 1919 the Hungarian engineer **Karl Ereky** first used the term "biotechnology", and in 1923 Pfizer started producing citric acid with *Aspergillus niger*.

The discovery of **penicillin** by Alexander Fleming in 1928, and its development by Florey and Chain during the Second World War, is the most important single event in the history of industrial microbiology. To produce penicillin on a large scale, scientists at Peoria in the USA developed **deep-tank submerged fermentation**: the fungus was grown inside large, stirred, aerated tanks instead of on the surface of shallow flasks. They also found that corn steep liquor greatly increased yield and discovered the high-yielding strain *Penicillium chrysogenum* NRRL 1951. The techniques developed for penicillin (sterile aeration, agitation, aseptic operation of large fermenters, and strain improvement by mutation) became the basis of the whole modern fermentation industry. Soon after, Selman Waksman discovered **streptomycin** (1944), and a "golden age" of antibiotic discovery followed.

After 1950 the field expanded to amino acids (glutamic acid by *Corynebacterium glutamicum*, 1957), microbial transformation of steroids, industrial enzymes and single cell protein. The discovery of the structure of DNA by **Watson and Crick** in 1953 and of restriction enzymes around 1970 led to the first recombinant DNA experiments by **Cohen and Boyer** in 1973. This marks the beginning of **modern biotechnology**, in which the genes of microorganisms are directly changed. In 1977 human somatostatin was produced in *E. coli*, and in 1982 recombinant human insulin became the first genetically engineered drug. Today microbial biotechnology uses genomics, metabolic engineering and genome editing to design microorganisms for specific purposes.

2.1 Eras of biotechnology

Era	Period	Features
Ancient biotechnology	Before 2000 BC (8000–4000 BC)	Domestication of plants and animals; fermented foods made by chance without knowing microbes
Classical biotechnology	2000 BC to 1900s AD	Deliberate fermentation for food and medicine; later scientific understanding of microbes (Leeuwenhoek, Pasteur, Koch)
Industrial / antibiotic era	1900–1953	Acetone–butanol, citric acid, penicillin, streptomycin; deep-tank fermentation; genetics

Era	Period	Features
Molecular era	1953–1976	DNA structure, genetic code, restriction enzymes
Modern biotechnology	1977 onwards	Genetic engineering, recombinant proteins, PCR, genomics, metabolic engineering

2.2 Ancient and classical period

- **About 6000 BC**: Sumerians and Babylonians brewed **beer** with yeast.
- **About 4000 BC**: Egyptians made **leavened bread** using yeast; wine, vinegar and cheese were also made. Cheese began among nomadic tribes about 4000 years ago using **rennet** from calf stomach.
- Food was preserved by drying, salting, smoking and by fermentation (the acid formed stops spoilage).
- **1673–1676**: **Antonie van Leeuwenhoek**, with his single-lens microscope, first described bacteria and protozoa ("animalcules").
- **1701**: Giacomo Pylarini practised inoculation against smallpox; **1796**: **Edward Jenner** used cowpox to vaccinate against smallpox.
- **1809**: **Nicolas Appert** developed canning (heat sterilization of food in sealed containers).

2.3 The golden age of microbiology (1850–1900)

- **Louis Pasteur** (1822–1895), "father of industrial microbiology": showed in **1857** that lactic acid fermentation and later alcoholic fermentation are caused by **living microorganisms**; different microbes give different products; solved "diseases" of wine and beer; introduced **pasteurization**; described the **Pasteur effect** (oxygen inhibits fermentation); disproved spontaneous generation (swan-neck flask, 1861).
- **Robert Koch**: pure culture techniques on solid media and Koch's postulates (1876–1884), which made pure-culture industrial fermentation possible.
- **1877**: Wilhelm Kühne coined the word **"enzyme"** (Greek, "in yeast").
- **1897**: **Eduard Buchner** showed that a **cell-free yeast extract** ("zymase") could ferment sugar to alcohol. This proved fermentation is due to **enzymes**, not the "vital force" of living cells; Nobel Prize 1907.

2.4 Industrial fermentation era (1900–1950)

- **1914–1918**: **Chaim Weizmann** developed the **acetone–butanol–ethanol (ABE) fermentation** using *Clostridium acetobutylicum* on starch (maize). Acetone was needed to make **cordite** (explosive) during World War I. This was the first large-scale industrial fermentation needing strict aseptic conditions.
- **1917–1923**: J. N. Currie showed *Aspergillus niger* makes **citric acid** in surface culture; **Pfizer** started commercial production in 1923.
- **1919**: **Karl Ereky** coined "biotechnology".
- **1928**: **Alexander Fleming** noticed that *Penicillium notatum* killed *Staphylococcus* colonies and named the substance penicillin.
- **1938–1941**: **Howard Florey and Ernst Chain** (Oxford) purified penicillin and showed its clinical value (Nobel 1945 with Fleming).
- **1941–1945**: At the **Northern Regional Research Laboratory, Peoria (USA)**, penicillin production moved from surface flasks to **submerged deep-tank fermentation**: **corn steep liquor** raised yield, and the high-yielding strain *P. chrysogenum* **NRRL 1951** was found on a mouldy cantaloupe. This created modern fermentation technology (aeration, agitation, sterile air).
- **1944**: **Selman Waksman**, Albert Schatz and Elizabeth Bugie discovered **streptomycin** from *Streptomyces griseus*, the first drug effective against tuberculosis (Nobel 1952). Waksman coined the word "antibiotic".
- **1947–1950**: chloramphenicol (*S. venezuelae*), chlortetracycline (1948, Duggar, *S. aureofaciens*), and other antibiotics: "golden age of antibiotics".

2.5 1950–1976

- **1950s**: microbial **biotransformation of steroids** (e.g. 11α-hydroxylation of progesterone by *Rhizopus*, 1952) made cortisone cheap.
- **1953**: **Watson and Crick** proposed the double-helix structure of DNA.
- **1957**: **Kinoshita** (Japan) produced **glutamic acid** with *Corynebacterium glutamicum*: start of amino acid fermentation.
- **1960s–1970s**: **single cell protein** (SCP) from hydrocarbons and methanol (ICI "Pruteen" with *Methylophilus methylotrophus* in a large airlift fermenter); industrial enzymes such as glucose isomerase.
- **1969**: first industrial use of an **immobilized enzyme** (aminoacylase, Tanabe Seiyaku, Japan, I. Chibata).
- **1970**: discovery of **restriction enzymes** (Hamilton Smith); **1972**: Paul Berg made the first recombinant DNA molecule; **1973**: **Cohen and Boyer** cloned DNA in *E. coli* using a plasmid.

2.6 Modern biotechnology (1977 onwards)

Year	Event
1975	Köhler and Milstein: hybridoma technique for monoclonal antibodies
1976	Genentech founded (first genetic engineering company)
1977	Genentech produced **somatostatin** in *E. coli*: first human protein from a cloned gene
1978	Human **insulin** gene expressed in *E. coli*; rat insulin made by rDNA at Harvard
1982	**Humulin** (recombinant human insulin) approved: first rDNA drug
1983	Kary Mullis invented **PCR**
1986	First recombinant vaccine: **hepatitis B** surface antigen made in yeast; India set up the **Department of Biotechnology (DBT)**
1990–2003	Human Genome Project
1995	First complete bacterial genome: *Haemophilus influenzae*
1998	Remicade (monoclonal antibody for Crohn's disease) approved
2012 onward	CRISPR-Cas9 genome editing (Doudna and Charpentier); synthetic biology and metabolic engineering

Indian milestones: Hindustan Antibiotics Ltd., Pimpri (1954) for penicillin; IDPL (1961); DBT (1986); India is now a major producer of vaccines and generic antibiotics.

Exam point: Write history as eras with names and years. Must-know: Leeuwenhoek 1676, Pasteur 1857, Buchner 1897, Weizmann WWI, Ereky 1919, Fleming 1928, Florey–Chain and Peoria 1940s, Waksman 1944, Watson–Crick 1953, Cohen–Boyer 1973, somatostatin 1977, insulin 1982.

3. TYPES OF BIOREACTORS

Detailed explanation

The bioreactor, or fermenter, is the heart of any industrial fermentation. Its job is to provide the microorganism with the best possible environment to grow and make product, while keeping out all other organisms. Inside the vessel, the organism needs the right temperature, the right pH, enough nutrients and, for aerobic organisms, a continuous supply of dissolved oxygen. At the same time, the heat produced by metabolism and by stirring must be removed, carbon dioxide must escape, foam must be controlled, and all of this must be done without allowing a single contaminating organism to enter, sometimes for more than a week.

The most difficult requirement is usually the **supply of oxygen**. Oxygen dissolves very poorly in water (only about 7–8 mg per litre at 30°C), and a dense culture of aerobic microbes can use this in a few seconds. Oxygen must therefore be supplied continuously by blowing sterile air into the vessel through a **sparger** and breaking the air into very small bubbles, so that the surface area for transfer of oxygen from gas to liquid is as large as possible. The rate of transfer is described by the volumetric mass transfer coefficient $\mathbf{k_L a}$; a well-designed fermenter has a high $k_L a$. In a stirred tank, the impeller breaks the bubbles and spreads them through the broth, while the baffles prevent the liquid from simply swirling round in a vortex and make the mixing effective.

Different types of bioreactor solve the problems of mixing and aeration in different ways. The **stirred tank** uses a motor-driven impeller; it gives excellent mixing and oxygen transfer and is used for most industrial processes, but it uses a lot of power and the high shear forces near the impeller can damage delicate cells such as animal cells and some fungal mycelia. The **bubble column** has no impeller at all; the rising air bubbles alone mix and aerate the medium, which makes it simple, cheap and gentle, but mixing is poorer. The **airlift** fermenter improves on the bubble column by dividing the vessel into a riser and a downcomer with a draft tube. Air is blown only into the riser, so the liquid there is lighter and rises; at the top the gas escapes, and the heavier, gas-free liquid sinks down the downcomer. This creates a regular circulation with low shear and low energy use, which is why airlift fermenters are used for shear-sensitive cells and for very large processes such as single cell protein production.

Some bioreactors are designed for **immobilized** cells or enzymes. In a **packed bed** reactor the biocatalyst is fixed on particles packed in a column, and the medium flows through the bed. This gives a high concentration of catalyst and continuous operation, but the bed can clog, and oxygen supply and mixing are poor. A **fluidized bed** reactor solves the clogging problem by pumping the liquid upward fast enough to keep the particles suspended; its expanded top slows the liquid so that the particles are not washed out. **Membrane or hollow fibre** reactors keep the cells on one side of a semi-permeable membrane; nutrients diffuse in and products diffuse out, so very high cell densities can be reached with almost no shear. This makes them ideal for animal cells producing monoclonal antibodies. Finally, **photobioreactors** are designed for photosynthetic organisms such as algae and cyanobacteria; their main design problem is to supply light evenly to all the cells, which is done by using thin transparent tubes or flat panels, or shallow open ponds.

In choosing a bioreactor, the engineer considers the oxygen demand of the organism, its sensitivity to shear, whether it is free or immobilized, the viscosity of the broth, the need for light, the scale of production and the cost. There is no single best bioreactor; each type is suited to a particular kind of process.

3.1 Definition and functions

A **bioreactor** or **fermenter** is a vessel in which microorganisms, animal or plant cells, or enzymes are grown or used under **controlled, aseptic** conditions to carry out a biochemical process. Its function is to give the best environment (temperature, pH, dissolved oxygen, nutrients, mixing) so that the organism forms the maximum amount of product.

3.2 Requirements of an ideal bioreactor

- Able to run **aseptically** for long periods (sterilizable by steam in place).
- Good **aeration and agitation** to meet oxygen demand, without damaging cells (low shear).
- Low power consumption; **temperature control** (jacket or coils); **pH control**.
- Sampling, feeding and harvesting facilities; minimum evaporation loss.

- Foam control; easy cleaning and maintenance; made of **non-toxic, corrosion-resistant** material (stainless steel grade 316, glass for small units).
- Can be scaled up.

3.3 Parts of a stirred tank fermenter

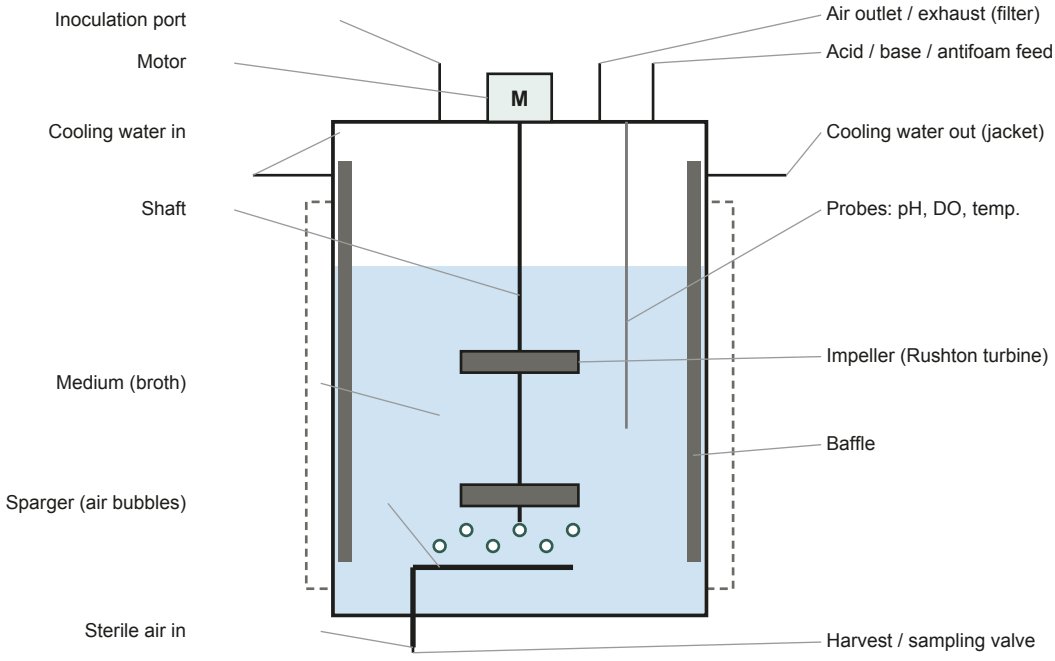


Fig. 2. Labelled diagram of a continuous stirred tank (stirred tank) fermenter

Part	Description and function
Vessel	Stainless steel cylinder with dished top and bottom; height : diameter about 2–3 : 1; working volume about 75–80% (headspace for foam)
Agitator (impeller)	Mounted on a shaft driven by a motor. **Rushton disc turbine** (flat blades on a disc) gives radial flow and breaks bubbles; marine propeller and pitched-blade turbines give axial flow. Mixes cells, nutrients and air and improves O_2 transfer
Sparger	Introduces sterile air at the bottom, below the impeller. Types: **porous** (sintered), **orifice** (perforated pipe or ring) and **nozzle** (open pipe)
Baffles	Usually **4 vertical metal strips**, width about 1/10 of vessel diameter, fixed to the wall; prevent **vortex** formation and improve mixing and aeration
Jacket / coils	Cooling or heating water circulates to keep temperature constant (removes metabolic and agitation heat)
Aeration system	Air compressor → sterile air filter → sparger; exhaust air through filter/condenser. Oxygen transfer is measured as $k_L a$
Foam control	Foam sensor with automatic addition of **antifoam** (silicone oils, vegetable oils) or mechanical foam breaker
Control probes	Temperature (thermocouple), **pH electrode** with acid/base pumps, **dissolved oxygen (DO) electrode**, pressure gauge, level sensor
Ports and valves	Inoculation, sampling, feeding (nutrients, precursor), and harvest valve; steam lines for sterilization in place

3.4 Types of bioreactors

(a) Continuous stirred tank reactor (CSTR)

- **Construction:** cylindrical vessel with motor-driven central shaft carrying one or more impellers, baffles, sparger, jacket (Fig. 2). Number of impellers depends on height.
- **Working:** air from the sparger is broken into small bubbles by the impeller and spread through the broth, giving a **uniform** environment.
- **Advantages:** efficient gas transfer, good mixing, flexible, easy control; the most widely used type.

- **Disadvantages:** high power use, high **shear** can damage filamentous fungi and animal cells, sealing of the shaft is a contamination risk.
- **Uses:** antibiotics, enzymes, amino acids, most aerobic fermentations.

(b) Bubble column bioreactor

- **Construction:** tall cylindrical column (height : diameter over 3 : 1) with a sparger at the base; **no mechanical agitator**; perforated plates may be placed inside to redistribute bubbles.
- **Working:** rising air bubbles both aerate and mix the medium.
- **Advantages:** simple, cheap, low shear, no moving parts, low contamination risk, low energy.
- **Disadvantages:** poor mixing at low gas flow, foaming, not good for viscous broth.
- **Uses:** baker's yeast, beer, vinegar, SCP, wastewater.

(c) Airlift bioreactor

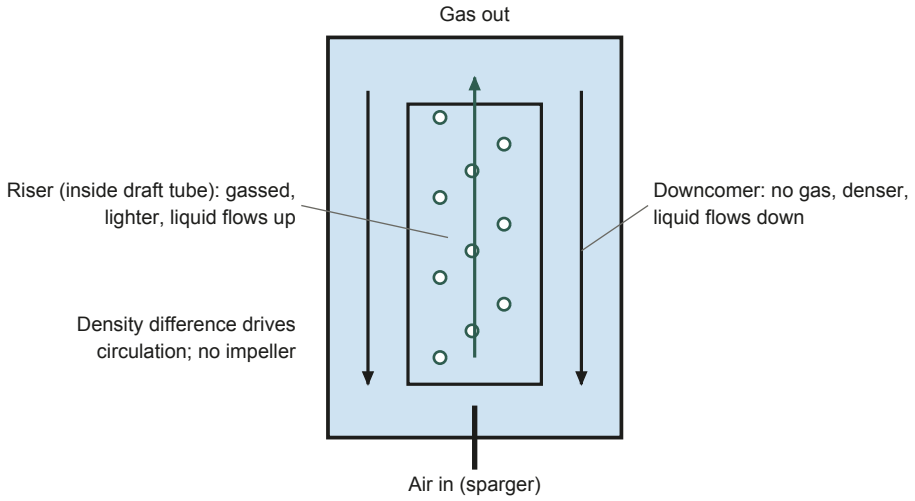


Fig. 3. Internal-loop airlift bioreactor (riser and downcomer)

- **Construction:** the vessel is divided into two zones by a **draft tube** or baffle: the **riser** (air is sparged into it) and the **downcomer** (no air). **Internal-loop** type has the draft tube inside; **external-loop** type has a separate outside tube connecting top and bottom.
- **Working:** the gassed liquid in the riser is less dense and rises; at the top gas escapes; the denser degassed liquid falls in the downcomer. This **density difference** causes circulation without any impeller.
- **Advantages:** low shear (good for plant and animal cells, fungi), low energy, good mixing and O_2 transfer, simple, easy to scale up, low contamination risk.
- **Disadvantages:** needs high air flow, less suitable for very viscous broths, foaming.
- **Uses:** SCP (ICI Pruteen plant, about 1500 m^3), plant and animal cell culture, monoclonal antibodies, wastewater treatment.

(d) Fluidized bed bioreactor

- **Construction:** similar to a bubble column but with a **narrow reaction column** and an **expanded top** section where fluid velocity falls, so particles fall back and are retained.
- **Working:** medium is pumped **upward** through a bed of small particles carrying immobilized cells or enzymes, fast enough to keep them **suspended (fluidized)**; liquid is often recycled.
- **Advantages:** good mixing and mass transfer, no clogging (unlike packed bed), suits immobilized cells, continuous operation.
- **Disadvantages:** hard to design and scale up, particles can be damaged.
- **Uses:** wastewater treatment (anaerobic), beer and vinegar with immobilized cells, immobilized enzyme reactions.

(e) Packed bed bioreactor

- **Construction:** a column packed with a **bed of solid particles** (porous or non-porous, rigid or compressible gels) with **immobilized** enzymes or cells on or inside them.
- **Working:** nutrient broth flows continuously through the bed (up or down); product leaves with the outflow. A **trickle bed** is a variant where liquid trickles down over the packing and air flows up.
- **Advantages:** high catalyst density, simple, continuous, high conversion, no need to separate catalyst from product.
- **Disadvantages:** poor mixing and O_2 supply, channelling, clogging, pH and heat gradients, compression of soft gels.
- **Uses:** immobilized glucose isomerase (HFCS), vinegar (trickling generator), effluent treatment.

(f) Membrane / hollow fibre bioreactor

- **Construction:** bundles of hollow semi-permeable fibres inside a shell. Cells are kept in the **shell (extracapillary) space**; medium flows through the fibres.
- **Working:** nutrients and oxygen diffuse out of the fibres to the cells; small products and wastes diffuse back into the flowing medium, while cells and large molecules are retained.
- **Advantages:** very high cell density, low shear, product separated continuously, cells protected.
- **Disadvantages:** membrane fouling, nutrient gradients, difficult scale-up, costly.
- **Uses:** animal cells, hybridomas (monoclonal antibodies), enzyme reactors, artificial organs.

(g) Photobioreactor

- **Definition:** a bioreactor with a **light source** for growing **photosynthetic** organisms (microalgae, cyanobacteria).
- **Open systems: raceway ponds** (shallow oval channels with a paddle wheel), cheap but easily contaminated, with poor control.
- **Closed systems: tubular** (transparent glass or plastic tubes), **flat-plate (panel)** and **column** reactors; good control of light, temperature, CO_2, and low contamination, but costly and can overheat.
- **Important factors:** light intensity and path (thin layers), CO_2 supply, O_2 removal (excess O_2 inhibits growth), temperature, mixing.
- **Uses:** *Spirulina*, *Chlorella* (SCP, health food), β-carotene (*Dunaliella*), astaxanthin (*Haematococcus*), algal biofuel.

3.5 Comparison of bioreactors

Bioreactor	Mixing by	Shear	Best for
Stirred tank (CSTR)	Impeller	High	Most bacterial and fungal fermentations
Bubble column	Rising air	Low	Yeast, simple aerobic processes
Airlift	Air + density difference (riser/downcomer)	Low	Shear-sensitive cells, SCP
Fluidized bed	Upward liquid flow	Medium	Immobilized cells, wastewater
Packed bed	Plug flow through bed	Low	Immobilized enzymes
Hollow fibre	Diffusion across membrane	Very low	Animal cells, antibodies
Photobioreactor	Air/pump + light	Low	Algae, cyanobacteria

Exam point: Always draw the stirred tank fermenter with at least 10 labels. For each other type write construction, working, advantages, disadvantages and uses. Airlift: remember riser, downcomer, draft tube, internal vs external loop.

4. TYPES OF FERMENTATION METHODS

Detailed explanation

Fermentations can be carried out in different ways depending on how nutrients are supplied and products removed, and on whether the medium is liquid or solid. The choice of method affects the yield, the productivity (amount of product per litre per hour), the cost and the risk of contamination.

In **batch fermentation** the fermenter is filled with medium, sterilized and inoculated, and then left to run until the end, with nothing added except air and pH or foam control agents. It is a **closed system**. As the organism grows, it passes through the familiar phases of the growth curve. In the **lag phase** the cells adapt to the new medium and synthesize the enzymes they need; there is little increase in number. In the **log or exponential phase** the cells divide at their maximum rate for the conditions, and the number doubles at regular intervals. As nutrients are used up and waste products accumulate, growth slows and the culture enters the **stationary phase**, where the number of cells remains constant. Finally, in the **death phase**, the cells die and lyse. Because conditions keep changing during a batch run, product formation depends on the phase: primary metabolites are formed in the log phase and secondary metabolites mainly in the stationary phase. Batch fermentation is simple and has a low risk of contamination, but time is lost between batches for emptying, cleaning, sterilizing and refilling.

Fed-batch fermentation is a batch process in which fresh nutrient is added gradually during the run while nothing is removed until the end. The main reason for feeding is to keep the concentration of the substrate low. High sugar concentrations can inhibit growth (substrate inhibition), can repress the enzymes needed for secondary metabolism (catabolite repression), and, in yeast, can cause ethanol formation instead of biomass (Crabtree effect). By feeding sugar slowly, the organism receives just enough to remain active, so production continues for much longer and higher yields are obtained. Toxic precursors, such as phenylacetic acid in penicillin production, can also be supplied safely in small portions. Fed-batch is the most widely used mode for antibiotics, baker's yeast and many recombinant proteins.

In **continuous fermentation** fresh medium flows into the fermenter at a constant rate and culture broth flows out at the same rate, so the volume stays the same. After a while the culture reaches a **steady state** in which the concentrations of cells, substrate and product no longer change. The most common continuous system is the **chemostat**, in which one nutrient is kept limiting. The rate of flow divided by the volume is called the dilution rate (D). At steady state, the cells must grow just fast enough to replace those washed out, so the specific growth rate equals the dilution rate ($\mu = D$). This means the growth rate can be controlled simply by adjusting the pump. If the dilution rate is set higher than the maximum growth rate of the organism, cells are removed faster than they can grow and the culture is **washed out**. Continuous culture gives the highest productivity and a uniform product, but running a fermenter for weeks without contamination is difficult, and improved strains tend to lose their special properties by mutation. It is therefore used mainly for simple, growth-linked products such as single cell protein, ethanol and waste treatment.

Fermentations are also classified by the physical state of the medium. In **submerged fermentation** the organism grows inside a liquid medium, which is aerated and stirred; this is easy to control and scale up and is used for most modern industrial processes. In **solid state fermentation** the organism grows on moist solid particles, such as wheat bran or rice straw, with little or no free water. It is similar to the natural growth of fungi on plant material, and it is well suited to filamentous fungi. It uses cheap agricultural residues, needs little water and energy, gives a concentrated product and has a low risk of bacterial contamination, but it is difficult to control temperature, moisture and pH, and difficult to scale up. The Japanese koji process for soy sauce and many fungal enzyme processes use solid state fermentation. **Semi-solid fermentation** lies between the two, using a thick slurry of solid substrate with some free water; it is often used for enzymes such as lipase.

Fermentations are classified by **mode of operation** (batch, fed-batch, continuous), by the **physical state of the medium** (submerged, surface, solid state, semi-solid), and by **oxygen requirement** (aerobic, anaerobic).

4.1 Batch fermentation (closed system)

- All nutrients are added at the start; the medium is sterilized and inoculated, and the fermentation runs to the end **without adding or removing** anything except air, acid/base for pH control and antifoam. At the end the whole broth is harvested.

- Cells pass through the **lag, log, stationary and death phases** (Fig. 1). Nutrients fall and products and waste build up, so conditions keep changing.
- **Kinetics:** in the log phase $dX/dt = \mu X$, where μ is the specific growth rate. Doubling time $t_d = 0.693/\mu$. The **Monod equation** relates μ to substrate: $\mu = \mu_{max} \cdot S / (K_s + S)$.
- **Advantages:** simple, less risk of contamination and strain mutation, flexible (one plant for many products), suits **secondary metabolites** made in stationary phase.
- **Disadvantages:** "down-time" between batches for cleaning, sterilizing and filling; low overall productivity; batch-to-batch variation.
- **Examples:** beer, wine, many antibiotics, enzymes, vaccines.

4.2 Fed-batch fermentation

- A batch culture that is **fed** with fresh nutrient (or precursor) continuously or at intervals during the run, **without removing broth**; the volume increases, and everything is harvested at the end.
- **Why used:** keeps substrate at a low level and so avoids **substrate inhibition** (e.g. high glucose), **catabolite repression** (glucose repressing secondary metabolism) and the **Crabtree effect** in yeast (ethanol formed at high sugar); allows toxic precursors (phenylacetic acid) to be added slowly; extends the production phase; gives high cell density.
- **Types:** variable-volume (volume increases) and fixed-volume (substrate fed as concentrated solution); feeding can be constant, exponential, or controlled by DO or pH (feedback).
- **Examples: penicillin**, baker's yeast, many recombinant proteins, amino acids.

4.3 Continuous fermentation (open system)

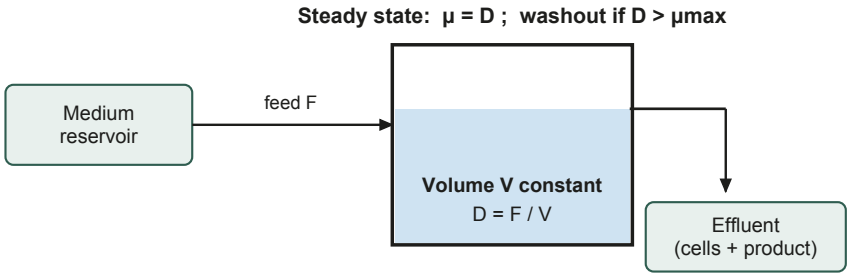


Fig. 4. Chemostat: fresh medium enters and broth leaves at the same rate

- Fresh medium is added **continuously** and culture broth is removed **at the same rate**, so the volume stays constant. After some time the culture reaches a **steady state** where cell, substrate and product concentrations stay constant and cells stay in the log phase.
- **Dilution rate** $D = F/V$ (F = flow rate, V = volume); units h^{-1}.
- Cell balance: $dX/dt = \mu X - DX$. At steady state $dX/dt = 0$, so **$\mu = D$**. The growth rate can therefore be set by the pump.
- If D is raised above μ_{max} (critical dilution rate), cells are removed faster than they grow and are lost: **washout**.

Type	Control	Notes
Chemostat	Constant flow of medium with one **growth-limiting nutrient**; growth rate set by D	Most common; stable at D below μ_{max}; used to study kinetics and physiology
Turbidostat	Cell density (turbidity) kept constant by a photocell that controls the feed pump	Works near μ_{max}; needs complex control
Multi-stage	Two or more vessels in series	First for growth, next for product formation (secondary metabolites)
With cell recycle	Cells separated (settling, membrane) and returned	Higher cell density and productivity; used in waste treatment and ethanol

- **Advantages:** high productivity, less down-time, uniform product, easier automation, cells kept in the most active state.

- **Disadvantages:** high risk of **contamination** over long runs, genetic instability and **back-mutation** of improved strains, not suitable for most secondary metabolites, wall growth, and difficult with filamentous organisms.
- **Examples:** SCP (Pruteen), ethanol, baker's yeast (some plants), vinegar, activated sludge process.

Feature	Batch	Fed-batch	Continuous
System	Closed	Semi-closed	Open
Volume	Constant	Increases	Constant
Nutrient addition	Only at start	During run	Continuous
Product removal	At end	At end	Continuous
Growth phase	All phases	Extended log / production	Steady log phase
Contamination risk	Low	Medium	High
Productivity	Low	High	Highest
Example	Beer, antibiotics	Penicillin, baker's yeast	SCP, ethanol

4.4 Submerged fermentation (SmF)

- Microorganisms grow **dispersed in a liquid medium** with aeration and agitation, in a stirred tank or airlift fermenter.
- **Advantages:** easy control of pH, temperature, O_2; uniform conditions; easy scale-up and automation; used for most industrial products.
- **Disadvantages:** high energy and water use, product is diluted, more effluent.
- **Examples:** antibiotics, most enzymes, citric acid (modern), vitamins, amino acids.

4.5 Surface (static) fermentation

- Organism grows as a **mat or pellicle on the surface** of a still liquid medium in shallow trays or pans; no agitation.
- **Examples:** early citric acid production by *A. niger* in aluminium trays, the Orleans method of vinegar, early penicillin in flasks.
- Now largely replaced by submerged fermentation because of low productivity and large space needs.

4.6 Solid state fermentation (SSF)

- Growth of microorganisms on **moist solid substrates** in the **absence or near absence of free-flowing water**. The solid gives both support and nutrients.
- **Substrates:** wheat bran, rice bran, rice straw, sugarcane bagasse, corn cobs, fruit pulp, oil cakes.
- **Organisms:** mainly **filamentous fungi** (*Aspergillus*, *Rhizopus*, *Trichoderma*, *Penicillium*), as they tolerate low water; some yeasts and bacteria.
- **Important factors:** moisture content (about 40–80%) and **water activity** (a_w), particle size, aeration, temperature (heat build-up is the main problem), pH, inoculum.
- **SSF bioreactors:** tray, packed bed (column), rotating drum, stirred and fluidized-bed reactors.
- **Examples: koji** (*Aspergillus oryzae* on rice/soybean for soy sauce and sake), tempeh (*Rhizopus*), mushroom cultivation, composting, enzymes (amylase, cellulase, pectinase, protease), gibberellic acid, fungal spores for biopesticides.

4.7 Semi-solid fermentation

- Intermediate between SmF and SSF: the substrate is a **thick slurry or paste** that has some free water but not enough to be a true liquid (e.g. bran with a higher water ratio).
- Combines some advantages of both: better mixing and heat removal than SSF, higher product concentration than SmF.
- Used for enzymes such as **lipase** and protease (syllabus practical: lipase by semi-solid fermentation).

Feature	Submerged (SmF)	Solid state (SSF)
Free water	Plenty	Little or none
Organisms	Bacteria, yeast, fungi	Mainly filamentous fungi
Aeration	Needs forced aeration and agitation	Air diffuses through spaces between particles
Control of pH, temperature	Easy and uniform	Difficult; gradients and heat build-up
Contamination	Bacterial contamination possible	Less (low water activity)
Product concentration	Low (dilute)	High (concentrated)
Energy, water, effluent	High	Low
Scale-up	Easy	Difficult
Examples	Penicillin, citric acid, enzymes	Koji, fungal enzymes, compost, mushrooms

4.8 Other classifications

- **Aerobic** (need O_2): antibiotics, citric acid, SCP, enzymes, vinegar. **Anaerobic**: ethanol, lactic acid, acetone–butanol, biogas.
- **Pure culture** vs **mixed culture** (curd, kefir, biogas, composting).

Exam point: Draw the growth curve and chemostat. Write μ = D at steady state and explain washout. Give the batch vs fed-batch vs continuous table and the SmF vs SSF table.

5. ENZYME PRODUCTION

Detailed explanation

Enzymes are proteins that act as biological catalysts. They speed up chemical reactions under mild conditions of temperature and pH, and they are highly specific for their substrates. Because of these properties they are widely used in industry to replace harsh chemical processes, in medicine as drugs, and in laboratories as analytical tools. Although enzymes can be obtained from plants and animals, more than 90% of industrial enzymes today come from microorganisms, mainly species of *Bacillus*, *Aspergillus*, *Trichoderma*, *Streptomyces* and yeasts.

Microbial enzymes are either **extracellular**, meaning that they are secreted into the medium, or **intracellular**, meaning that they remain inside the cell. Extracellular enzymes such as amylase, protease, lipase and cellulase are the easiest to produce, because after fermentation the cells are simply removed and the enzyme is present in the liquid. Intracellular enzymes, such as glucose isomerase and asparaginase, require the cells to be broken open first, which adds cost. Many enzymes are also **inducible**: the organism makes them only when their substrate (the inducer) is present. For this reason starch is added to the medium to produce amylase and oil to produce lipase. On the other hand, easily used sugars such as glucose often switch off enzyme synthesis (catabolite repression), so producers use slowly metabolized carbon sources or feed glucose gradually.

The production process begins with the selection of a suitable organism. It should give a high yield in a short time, be safe (preferably generally recognized as safe), not produce toxins or antibiotics, and grow on cheap media. The organism is grown up from a stock culture through shake flasks and seed fermenters to provide an active inoculum. Production is usually carried out by submerged fermentation for bacterial enzymes and often by solid state fermentation for fungal enzymes. Conditions such as temperature, pH and aeration are optimized for enzyme formation rather than for growth, and the fermentation is stopped when enzyme activity reaches its maximum, usually towards the end of the log phase or early stationary phase.

Recovery of the enzyme (downstream processing) must be gentle, because enzymes are easily denatured by heat, extreme pH and organic solvents. The cells are first removed by centrifugation or filtration. For intracellular enzymes the cells are broken by high-pressure homogenizers, bead mills, ultrasound or lytic enzymes. The dilute enzyme solution is concentrated, usually by ultrafiltration, and the enzyme may be precipitated with ammonium sulphate or with cold organic solvents. Industrial enzymes for detergents or starch processing need only partial purification, but enzymes for medical or analytical use are purified further by chromatography. Finally, the enzyme is formulated with stabilizers, standardized to a fixed activity, and sold as a liquid, powder or granules. The activity of an enzyme is expressed in units; one unit is the amount of enzyme that converts one micromole of substrate per minute under defined conditions.

Microbial enzymes have three main groups of applications. In **industry** they are used in detergents (proteases, lipases, amylases), starch processing (amylases, glucoamylase, glucose isomerase for high fructose syrup), baking, brewing, fruit juice clarification (pectinase), dairy (rennet, lactase), textiles, leather and paper. In **medicine** they are used as therapeutic agents, for example L-asparaginase for leukaemia and streptokinase to dissolve blood clots. In **analysis** they are used in diagnostic kits and biosensors, such as glucose oxidase in blood glucose strips, and as tools in molecular biology, such as restriction enzymes and Taq polymerase.

5.1 Why microbial enzymes

- Microbes grow fast, give **high yields** in days, and can be grown in any season.
- Many enzymes are **extracellular**, so extraction is easy and cheap.
- Large variety of enzymes, including ones stable at high temperature (thermophiles), extreme pH (alkaliphiles) or in detergents.
- Yield can be increased by strain improvement and genetic engineering; cheap media.

5.2 Types of microbial enzymes

- **Extracellular** (exoenzymes): secreted into the medium, e.g. amylase, protease, lipase, cellulase, pectinase. Easy to recover.

- **Intracellular** (endoenzymes): stay inside cells, e.g. glucose isomerase, glucose oxidase, asparaginase, penicillin acylase. Cells must be broken.
- **Constitutive**: always made. **Inducible**: made only when an **inducer** (usually the substrate) is present, e.g. amylase by starch, cellulase by cellulose.

5.3 Regulation that affects yield

- **Induction**: add the inducer to the medium.
- **Catabolite repression**: glucose represses many enzymes, so use slowly used carbon sources or fed-batch feeding.
- **Feedback (end-product) repression**: product accumulation lowers synthesis; use regulatory mutants.
- Improve yield by mutants that are **constitutive** and **catabolite-derepressed**, and by cloning extra gene copies.

5.4 Steps in industrial enzyme production

1. **Selection of organism**: high yield, GRAS, stable, short fermentation, no toxins or antibiotics; e.g. *Bacillus*, *Aspergillus*, *Trichoderma*, *Streptomyces*, yeasts.
2. **Medium**: cheap carbon (starch, molasses, bran), nitrogen (soybean meal, CSL, ammonium salts), minerals, **inducer**; pH and C:N ratio optimized.
3. **Sterilization** of medium and fermenter.
4. **Inoculum**: spores or cells → shake flasks → seed tank (about 5–10% v/v).
5. **Fermentation**: SmF (most bacterial enzymes, 2–7 days) or SSF (many fungal enzymes); controlled temperature, pH, aeration; harvest when activity is highest (usually late log or early stationary).
6. **Downstream processing** (Fig. 5): separation, cell disruption (for intracellular), concentration, precipitation, purification.
7. **Formulation and standardization**: add stabilizers (glycerol, sorbitol, NaCl, calcium), preservatives; make liquid, powder or dust-free granules (for detergents); adjust to a fixed activity.

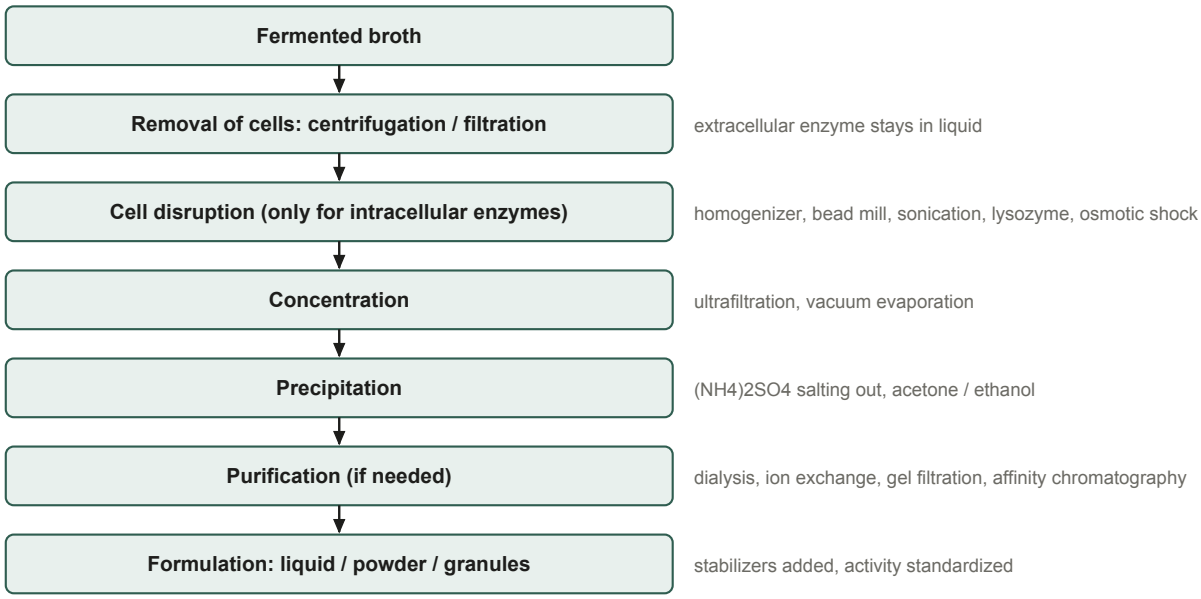


Fig. 5. Downstream processing of microbial enzymes

5.5 Examples of enzyme production

α-Amylase (bacterial)

- Organism: *Bacillus amyloliquefaciens*, *B. licheniformis* (thermostable, active at about 90°C), *B. subtilis*.
- Medium: starch (inducer), soybean meal or CSL, $CaCl_2$ (calcium stabilizes amylase); submerged, aerobic, about 30–40°C, pH about 7, 2–4 days.

- Recovery: remove cells, concentrate by ultrafiltration, precipitate or sell as stabilized liquid.
- Fungal amylase is made by *Aspergillus oryzae* in SSF (wheat bran) or SmF.
- Uses: starch liquefaction for glucose syrups, baking, brewing, textile desizing, detergents.

Protease (alkaline, detergent)

- Organism: *Bacillus licheniformis* (subtilisin Carlsberg), *B. subtilis*, alkalophilic *Bacillus*.
- Medium: starch, soybean meal, casein; submerged, aerobic, about 30–37°C, pH 7–9, 2–3 days.
- Recovery: filtration, concentration, precipitation, granulation (dust-free granules to avoid allergy).
- Uses: detergents (largest market), leather dehairing, meat tenderizing, silver recovery from X-ray film.

5.6 Enzyme activity and purity

- **Enzyme unit (U)**: the amount of enzyme that converts **1 µmol of substrate per minute** under defined conditions. SI unit: **katal** (1 mol/s).
- **Specific activity** = units per mg protein; rises during purification.
- **Purification fold** = specific activity after step ÷ specific activity of crude extract; **yield (%)** = total units recovered ÷ total units at start × 100.

5.7 Important microbial enzymes and applications

Enzyme	Source	Application
α-Amylase	*B. amyloliquefaciens*, *B. licheniformis*, *A. oryzae*	Starch liquefaction, baking, brewing, textile desizing, detergent
Glucoamylase	*Aspergillus niger*, *Rhizopus*	Glucose syrup, brewing, ethanol
Alkaline protease	*B. licheniformis*, *B. subtilis*	Detergents, leather, meat tenderizing
Lipase	*Candida rugosa*, *Rhizopus*, *Aspergillus*, *Pseudomonas*	Detergent, dairy flavour, fat modification, biodiesel
Cellulase	*Trichoderma reesei*, *Aspergillus*	Bioethanol, stone-washed jeans, juice extraction, feed
Pectinase	*Aspergillus niger*	Clarification of fruit juice and wine
Glucose isomerase	*Streptomyces*, *Actinoplanes*	High fructose corn syrup (HFCS)
Invertase	*Saccharomyces cerevisiae*	Invert sugar, soft-centre chocolates
Lactase (β-galactosidase)	*Kluyveromyces lactis*, *Aspergillus*	Lactose-free milk, whey use
Microbial rennet / chymosin	*Rhizomucor miehei*; recombinant in *K. lactis*, *A. niger*	Cheese making
Xylanase	*Trichoderma*, *Aspergillus*	Pulp bleaching, baking, feed
Penicillin acylase	*E. coli*, *Bacillus megaterium*	6-APA for semi-synthetic penicillins

Therapeutic enzymes

Enzyme	Source	Use
L-Asparaginase	*E. coli*, *Erwinia chrysanthemi*	Acute lymphoblastic leukaemia (removes asparagine needed by tumour cells)
Streptokinase	β-haemolytic *Streptococcus*	Dissolves blood clots (heart attack), activates plasminogen
Urokinase	Human kidney cells / recombinant *E. coli*	Thrombolytic (clot dissolving)
Serratiopeptidase	*Serratia marcescens*	Anti-inflammatory
Collagenase	*Clostridium histolyticum*	Wound debridement, burns
Lysozyme	Egg white / microbial	Antibacterial

Analytical enzymes

- **Glucose oxidase** (*A. niger*) + peroxidase: blood and urine glucose tests, glucose **biosensors** (glucometer strips).
- **Cholesterol oxidase** (*Streptomyces*, *Nocardia*): serum cholesterol test. **Urease**: blood urea. **Luciferase**: ATP detection (hygiene tests).
- **Horseradish peroxidase** and **alkaline phosphatase**: enzyme labels in **ELISA**.
- Molecular biology: **restriction endonucleases**, DNA ligase, **Taq polymerase** (*Thermus aquaticus*, PCR), reverse transcriptase.

Exam point: Answer order: why microbes → types (extra/intracellular, inducible) → steps with medium and conditions → downstream flow chart → one example (amylase or protease) → therapeutic / analytical / industrial applications table.

6. IMMOBILIZATION OF ENZYMES

Detailed explanation

A major disadvantage of using free (soluble) enzymes in industry is that they are used only once. After the reaction, the enzyme remains dissolved in the product and is either lost or must be removed, and it cannot easily be recovered for reuse. Since many enzymes are expensive, this makes the process costly. **Immobilization** solves this problem by attaching the enzyme to, or trapping it in, an insoluble material called a support or matrix. The immobilized enzyme can then be separated from the product simply by filtration or by keeping it in a column, and it can be used again and again, or continuously for weeks or months.

Immobilization has several further advantages. It allows the use of continuous reactors such as packed bed and fluidized bed reactors, in which the substrate solution flows over the immobilized enzyme and product comes out continuously. The product is not contaminated with enzyme protein, which simplifies purification. Many enzymes become more stable after immobilization, because the support protects them from unfolding at high temperature or extreme pH. The reaction can be stopped at any time by removing the enzyme. However, immobilization also has disadvantages: some enzyme activity is usually lost in the process, the support adds cost, and because the substrate has to diffuse to the enzyme through the support, the reaction may be slower (diffusion limitation), especially for large substrates.

There are five main methods of immobilization. In **adsorption**, the enzyme sticks to the surface of a support such as charcoal, silica or an ion-exchange resin by weak physical forces. This is the simplest and mildest method, but the enzyme may gradually come off the support. In **covalent bonding**, strong chemical bonds are formed between functional groups of the enzyme and a chemically activated support, for example agarose activated with cyanogen bromide. The enzyme cannot leak out, but the chemical reactions may damage the enzyme and reduce its activity. In **entrapment**, the enzyme or whole cells are trapped inside the network of a gel, most commonly calcium alginate. A mixture of sodium alginate and enzyme is dropped into calcium chloride solution, where the alginate forms firm beads with the enzyme inside. Small substrate molecules diffuse into the beads, but the enzyme cannot escape. In **encapsulation**, the enzyme is enclosed within a semi-permeable membrane, forming tiny capsules. In **cross-linking**, enzyme molecules are joined to each other by a bifunctional reagent such as glutaraldehyde, forming insoluble aggregates without any separate support.

Immobilized enzymes and cells are used in several major industrial processes. The most important is the production of **high fructose corn syrup** using immobilized glucose isomerase, which converts glucose into fructose, a sweeter sugar. Immobilized aminoacylase is used to separate L-amino acids from mixtures of D and L forms; this was the first industrial application of an immobilized enzyme (Tanabe, Japan, 1969). Immobilized penicillin acylase is used to make 6-aminopenicillanic acid, the starting material for semi-synthetic penicillins, and immobilized lactase is used to make lactose-free milk. Immobilized enzymes are also the basis of **biosensors**, such as the glucose electrode used by diabetic patients.

6.1 Definition

Enzyme immobilization is the attachment or confinement of an enzyme (or whole cells) on or within an inert, insoluble **support (matrix)** so that it keeps its catalytic activity and can be **used repeatedly or continuously**. The first example was **invertase adsorbed on charcoal** (Nelson and Griffin, **1916**); the first industrial process was **aminoacylase** on DEAE-Sephadex for L-amino acids (**Chibata, Tanabe Seiyaku, Japan, 1969**).

6.2 Advantages

- **Reuse** of the enzyme many times, which lowers cost.
- Allows **continuous** processes in packed bed or fluidized bed reactors.
- Product is **not contaminated** by enzyme, so purification is easy.
- Often **greater stability** to temperature, pH, organic solvents and storage.
- Reaction can be **stopped quickly** by removing the enzyme; better process control.
- High enzyme concentration per reactor volume.

6.3 Disadvantages

- **Loss of activity** during immobilization (active site blocked or changed).
- **Diffusion (mass transfer) limitation**: substrate must reach the enzyme; big substrates move slowly.
- Extra cost of support and process.
- Leakage of enzyme from some supports; microbial contamination in long runs.
- Not useful when the substrate is insoluble or very large.

6.4 Properties of a good support

- Inert, non-toxic, insoluble, cheap and easily available.
- Mechanically strong, chemically and microbially stable.
- Large surface area, porous, allowing substrate and product to diffuse.
- Has groups for binding (-OH, -NH_2, -COOH); can be regenerated.

Type of support	Examples
Natural organic polymers	Alginate, agar, agarose, carrageenan, cellulose, chitosan, gelatin, collagen, starch
Synthetic organic polymers	Polyacrylamide, polystyrene, nylon, polyvinyl alcohol, ion-exchange resins
Inorganic materials	Activated charcoal, porous glass, silica, alumina, clay, bentonite

6.5 Methods of immobilization

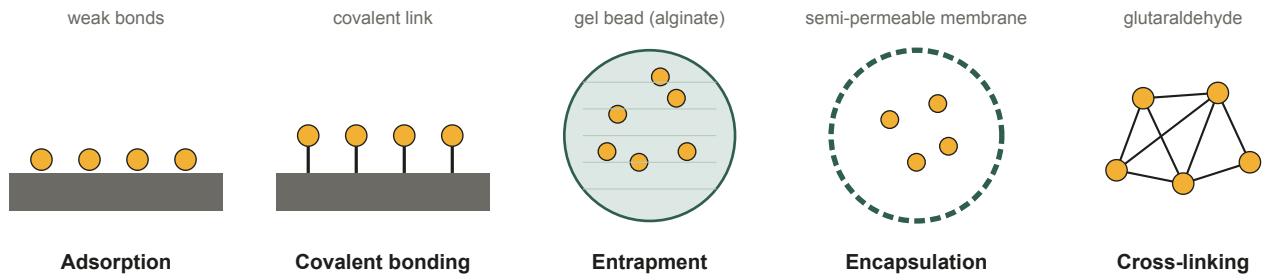


Fig. 6. Methods of enzyme immobilization (enzyme shown as yellow circles)

(1) Adsorption

- Enzyme is bound to the surface of the support by **weak forces**: van der Waals, ionic, hydrogen and hydrophobic bonds.
- **Procedure:** mix enzyme solution with the support under suitable pH and ionic strength, then wash off unbound enzyme.
- **Supports:** activated charcoal, alumina, silica, clay, glass, ion-exchange materials (DEAE-cellulose, CM-cellulose, DEAE-Sephadex).
- **Merits:** simple, cheap, mild, little change in enzyme, support can be reused. **Demerits:** enzyme **leaks (desorbs)** with change in pH, temperature or ionic strength.

(2) Covalent bonding

- Enzyme is joined by **covalent bonds** between its functional groups (-NH_2 of lysine, -COOH, -SH, -OH) and a **chemically activated** support. Groups needed for catalysis must not be used.
- **Activating agents:** cyanogen bromide (CNBr) for agarose/Sepharose and cellulose, **glutaraldehyde**, carbodiimide, silanes for glass.
- **Merits:** strong binding, **no leakage**, stable. **Demerits:** harsh chemistry can **inactivate** the enzyme; support cannot be reused; costly.

(3) Entrapment

- Enzyme or whole cells are **physically trapped** inside the lattice of a polymer gel or fibres; small substrates and products diffuse through the pores, the enzyme cannot escape.

- **Calcium alginate method (most common):** mix enzyme or cells with **2–3% sodium alginate** → drop through a syringe into **0.1–0.2 M $CaCl_2$** → Ca^{2+} cross-links alginate and forms **beads** → harden for 30 min, wash.
- Other gels: **polyacrylamide** (polymerized with bis-acrylamide), κ-carrageenan (with KCl), agar, gelatin, polyvinyl alcohol.
- **Merits:** mild, simple, suits **whole cells**, no chemical change of enzyme. **Demerits:** small enzymes can leak; diffusion limitation; alginate dissolves with phosphate or citrate (chelators).

(4) Encapsulation (microencapsulation)

- Enzyme is enclosed inside **semi-permeable membrane capsules** (microcapsules about 10–100 μm) or liposomes. Only small molecules pass through the membrane.
- **Membranes:** nylon, cellulose nitrate, polyethyleneimine; lipid vesicles (liposomes); hollow fibre membranes.
- **Merits:** large amount of enzyme held; several enzymes can be co-encapsulated. **Demerits:** only for small substrates; capsules may break.

(5) Cross-linking (carrier-free)

- Enzyme molecules are joined **to each other** by covalent bonds using **bifunctional reagents**, forming insoluble aggregates without any support.
- **Reagent: glutaraldehyde** (most common), hexamethylene diisocyanate; modern forms: CLEC (cross-linked enzyme crystals), **CLEA** (cross-linked enzyme aggregates).
- **Merits:** simple, no support cost, high enzyme density. **Demerits:** loss of activity, gel-like aggregates weak and hard to handle; often combined with adsorption.

Feature	Adsorption	Covalent	Entrapment	Encapsulation	Cross-linking
Binding force	Weak physical	Covalent	Physical trapping	Membrane barrier	Covalent (enzyme-enzyme)
Preparation	Simple	Difficult	Moderate	Moderate	Simple
Activity loss	Low	High	Low	Low	Moderate–high
Enzyme leakage	Yes	No	Possible	No (small enzymes may)	No
Cost	Low	High	Low	Moderate	Low

6.6 Changes in properties after immobilization

- **Stability** usually increases (thermal, pH, storage).
- **Optimum pH** may shift because of the charge of the support (microenvironment effect).
- **Apparent K_m** often increases because of diffusion limitation; V_{max} may fall.
- Activity towards large substrates falls (steric hindrance).

6.7 Immobilized whole cells

- Cells are immobilized (usually in alginate or carrageenan) when the enzyme is intracellular, unstable when purified, or needs cofactors.
- Advantages: no enzyme extraction, multi-step reactions possible, cofactors regenerated. Examples: immobilized *E. coli* (aspartase) for L-aspartic acid (Tanabe, 1973); *Brevibacterium* (fumarase) for L-malic acid; immobilized yeast for ethanol and beer; *Rhodococcus* (nitrile hydratase) for acrylamide.

6.8 Applications

Immobilized enzyme / cell	Product / use
Glucose isomerase (*Streptomyces*)	Glucose → fructose: **high fructose corn syrup**; the largest immobilized enzyme process
Aminoacylase (*Aspergillus oryzae*)	Resolution of DL-amino acids to L-amino acids (L-methionine, L-phenylalanine); first industrial process 1969

Immobilized enzyme / cell	Product / use
Penicillin G acylase (*E. coli*)	Penicillin G → **6-APA** for ampicillin, amoxicillin
Lactase (β-galactosidase)	Lactose-free milk; whey hydrolysis
Invertase	Invert sugar syrup
Nitrile hydratase (*Rhodococcus* cells)	Acrylonitrile → **acrylamide**
Aspartase (*E. coli* cells)	L-aspartic acid (for aspartame sweetener)
Glucose oxidase on electrode	**Biosensors** for blood glucose
Urease in microcapsules	Artificial kidney (removal of urea)

Exam point: Draw Fig. 6 and describe each method with principle, support/agent, merits and demerits. Mention the Ca-alginate bead procedure, glutaraldehyde, and glucose isomerase → HFCS as the main application.

7. SCREENING AND STRAIN IMPROVEMENT

Detailed explanation

Every industrial fermentation depends on a good microorganism. The organisms used today were originally isolated from nature, most often from soil, and then improved over many years. The process of finding a useful organism is called **screening**, and the process of making it better is called **strain improvement**.

Nature contains an enormous number of microorganisms, but only a very small fraction produce a given useful compound in useful amounts. Screening is therefore designed to search through thousands of isolates quickly and cheaply to find the few that are worth studying. **Primary screening** is the first, qualitative step: it simply detects which organisms make the product. For antibiotic producers the classic method is the **crowded plate technique**, in which soil is plated at a dilution that gives many colonies; a colony surrounded by a clear zone where other colonies cannot grow is probably producing an antibiotic. For enzymes, special plates reveal activity: a clear zone on starch agar after adding iodine shows amylase, a clear zone on skim milk agar shows protease, and so on. **Auxanography** is used to find organisms producing vitamins or amino acids, by looking for growth of a test organism that requires the compound. **Enrichment culture** favours organisms with a particular ability, for example by supplying a pollutant as the only carbon source.

Secondary screening is carried out on the small number of promising isolates selected by primary screening. It is quantitative and much more detailed. The organisms are grown in shake flasks or small fermenters to measure exactly how much product they make and how fast. The organism is identified, the product is checked to see whether it is new or already known, and the best conditions of pH, temperature, aeration and medium are worked out. Secondary screening also tells whether the organism is stable, whether it is safe, and whether the process is likely to be economical. Only organisms that pass secondary screening are developed further.

Wild organisms usually make very little product, because in nature their metabolism is tightly regulated so that they do not waste energy making more than they need. **Strain improvement** aims to remove this regulation and increase yield, and also to make the strain grow on cheaper media, produce fewer by-products and be easier to handle. The classical method is **mutation and selection**. Cells or spores are treated with a mutagen, such as ultraviolet light or the chemical NTG, which causes random changes in the DNA. Most survivors are no better or worse, but a few may produce more. Thousands of survivors are tested and the best are selected; the process is repeated many times, each time starting with the best strain. The classic example is penicillin: starting from Fleming's strain, which gave only a few units per millilitre, repeated rounds of mutation with X-rays, ultraviolet light and chemicals produced modern strains giving over 50,000 units per millilitre.

More directed methods are also used. Mutants that are resistant to toxic analogues of the product often have enzymes that are no longer inhibited by the product, so they overproduce it. Auxotrophic mutants, which are blocked at one step of a pathway, may accumulate the compound just before the block. **Recombination** methods, such as protoplast fusion and the parasexual cycle in fungi, combine desirable characters from different strains. Finally, **recombinant DNA technology** allows specific genes to be added, removed or altered: extra copies of key genes can be introduced, feedback controls can be removed, and genes from other organisms can be expressed, as in the production of human insulin in *E. coli*. Improved strains are valuable and must be preserved carefully, for example by freeze-drying or storage in liquid nitrogen, so that their properties are not lost.

7.1 Screening

Screening is the use of highly selective procedures to **detect and isolate** microorganisms of industrial interest from a large microbial population. It separates the few useful isolates from thousands of useless ones in the least time and cost. Sources are soil (richest source, especially for actinomycetes and fungi), water, sewage, compost, decaying food, extreme environments, and culture collections (ATCC, MTCC Chandigarh, NCIM Pune, NRRL).

7.2 Primary screening (qualitative)

Detects **which** organisms make the desired product. It is quick, simple and done on a large number of isolates, usually on agar plates.

(a) Crowded plate technique (for antibiotic producers)

- Soil is diluted and plated so that plates are **crowded** with 300–400 colonies.
- Colonies producing antibiotics show a **clear zone of inhibition** around them where other colonies do not grow.
- These colonies are picked and purified. The technique is simple but only detects producers active against neighbouring colonies.

(b) Test organism / cross-streak and agar plug methods

- **Crowded plate with test organism**: after growth the plate is overlaid with soft agar containing a test bacterium (e.g. *Staphylococcus aureus*); zones show antibiotic activity against it.
- **Cross-streak**: isolate streaked in one line; test organisms streaked at right angles after growth; inhibition near the isolate shows its **spectrum**.
- **Giant colony technique**: a single large colony is grown and tested against many organisms.

(c) Auxanography (for growth-factor producers)

- Used to find organisms that make **vitamins or amino acids**.
- Plate seeded with a test organism that **needs** that growth factor; isolates are spotted on it. A **zone of growth** of the test organism around an isolate shows it produces the growth factor.

(d) Indicator dye / plate assay methods

Product	Plate method
Organic acids	pH indicator dye (bromocresol green, bromothymol blue) or $CaCO_3$ in medium: yellow zone or clear zone of dissolved $CaCO_3$
Amylase	Starch agar flooded with iodine: clear zone on blue background
Protease	Skim milk or casein agar: clear zone
Lipase	Tributyrin agar: clear zone; or Rhodamine B–olive oil plate (orange fluorescence)
Cellulase	Carboxymethyl cellulose (CMC) agar stained with Congo red: clear zone
Phosphate solubilizers	Pikovskaya agar: clear halo

(e) Enrichment culture technique

- A medium is made that favours only the desired type, e.g. only the target compound as sole carbon source (for degraders), or selective temperature/pH (for thermophiles, alkaliphiles). Repeated subculturing enriches the wanted organism.

7.3 Secondary screening (quantitative)

Done on the few organisms selected in primary screening, usually in **shake flasks** or small fermenters. It tells **how much** product is made and whether the organism is worth developing.

- Measures **yield**, productivity and fermentation time; compares isolates.
- **Identifies** the organism (to check pathogenicity and whether the product is new or already known).
- Finds whether the product is new, and its chemical and biological properties (spectrum, toxicity).
- Determines optimum **pH, temperature, aeration, medium** components and the need for special conditions.
- Shows whether the product is intracellular or extracellular and whether the organism is genetically stable.
- Checks if the product is destroyed by the organism itself, and economics of the process.

7.4 Strain improvement

Wild isolates usually give very low yields. **Strain improvement** is the deliberate genetic change of a microorganism to make it better for industry.

Aims

- Increase **yield** and productivity; shorten fermentation time.
- Use of **cheaper** substrates; reduce unwanted by-products and foaming.

- Improve genetic stability, tolerance to product, and ease of downstream processing (e.g. extracellular product, pellet morphology).

(A) Mutation and selection (classical method)

Cells or spores are exposed to a **mutagen** to give about 90–99% kill; survivors are screened for improved mutants. Repeated rounds give stepwise improvement.

Mutagen	Type	Action
UV light (260 nm)	Physical	Forms **thymine dimers**; errors during repair (SOS) cause mutations
X-rays, γ-rays	Physical (ionizing)	Break DNA strands, cause deletions and rearrangements
NTG (N-methyl-N′-nitro-N-nitrosoguanidine)	Chemical, alkylating	Methylates guanine → GC to AT transitions; most potent, acts at replication fork
EMS (ethyl methane sulphonate)	Chemical, alkylating	Ethylates guanine → GC to AT transitions
Nitrous acid	Chemical, deaminating	Deaminates adenine and cytosine → transitions
Base analogues (5-bromouracil, 2-aminopurine)	Chemical	Incorporated in DNA and mispair
Acridine dyes (acridine orange, proflavine)	Chemical, intercalating	Insert between bases → frameshift mutations

Selection of mutants

- **Random screening**: test thousands of survivors in flasks for higher yield (used for penicillin).
- **Auxotrophic mutants**: need a growth factor; found by **replica plating** and enriched by penicillin enrichment. Block a pathway so an intermediate accumulates (e.g. homoserine auxotroph of *C. glutamicum* overproduces **lysine**).
- **Regulatory (analogue-resistant) mutants**: grown on a toxic **analogue** of the end product; resistant mutants often have enzymes insensitive to **feedback inhibition** and overproduce (e.g. AEC-resistant lysine producers).
- **Catabolite-derepressed** and **constitutive** mutants for enzymes; **revertants** of non-producing mutants that sometimes produce more.

(B) Recombination

- **Protoplast fusion**: cell walls removed by lysozyme (bacteria) or lytic enzymes (fungi) in an osmotic stabilizer (sucrose, mannitol, KCl) → protoplasts of two strains mixed with **polyethylene glycol (PEG)** → fusion → regeneration of cell wall on osmotically stabilized medium → selection of recombinants. Used for *Streptomyces*, *Penicillium*, yeasts.
- **Parasexual cycle** (Pontecorvo) in fungi without a sexual stage: heterokaryon → diploid → mitotic recombination → haploidization; used in *Aspergillus niger* and *P. chrysogenum*.
- **Transformation, transduction and conjugation** in bacteria; sexual hybridization in yeasts.

(C) Recombinant DNA technology and modern methods

- **Gene cloning**: extra copies of key genes, strong promoters, removal of repressors; expression of foreign genes in *E. coli*, *Bacillus*, yeast (insulin, chymosin, interferon).
- **Site-directed mutagenesis / protein engineering**: change a specific amino acid, e.g. oxidation-resistant subtilisin for bleach detergents.
- **Metabolic engineering**: block competing pathways and remove feedback control, e.g. riboflavin-overproducing *B. subtilis*.
- **Directed evolution**, **genome shuffling**, and genome editing (CRISPR).

7.5 Classic example: penicillin strain improvement

Strain	How obtained	Approx. yield
P. notatum (Fleming)	Original isolate, surface culture	About 2–20 units/mL

Strain	How obtained	Approx. yield
P. chrysogenum NRRL 1951	Natural isolate from a mouldy cantaloupe (Peoria)	About 100 units/mL (submerged)
NRRL 1951 B25	Single-spore selection	About 250 units/mL
X-1612	**X-ray** mutant	About 500 units/mL
Wis Q-176	**UV** mutant (University of Wisconsin); non-pigmented later mutants	About 900 units/mL
Modern industrial strains	Repeated mutation (NTG, UV) and selection	Over 50,000 units/mL (tens of g/L)

7.6 Preservation of industrial strains

- **Subculturing** on slants and storage at 4°C (short term); **overlaying with sterile mineral oil**.
- Storage as dried spores in **soil or silica gel** (fungi, actinomycetes).
- **Lyophilization (freeze-drying)**: cells frozen and dried under vacuum, stored for years.
- **Cryopreservation** in **liquid nitrogen (−196°C)** or at −80°C with glycerol or DMSO as cryoprotectant (best for long-term stability).

Exam point: Screening: define, then primary (crowded plate, auxanography, indicator plates, enrichment) and secondary (list 6 points). Strain improvement: aims → mutagens table → mutant selection → protoplast fusion → rDNA → penicillin example.

8. ANTIBIOTIC PRODUCTION

Detailed explanation

Antibiotics are chemical substances produced by microorganisms that, in small amounts, inhibit the growth of or kill other microorganisms. They are secondary metabolites, which means that the producing organism makes them mainly after its rapid growth phase is over, and they are not essential for its own growth. In nature they probably help the producer compete with other microbes in the soil. Most antibiotics come from actinomycetes, especially the genus *Streptomyces*, and from filamentous fungi such as *Penicillium* and *Cephalosporium*.

Industrial antibiotic production follows a general pattern. A highly improved strain, kept in a stable form such as freeze-dried spores, is grown up through several inoculum stages. Production takes place in large stirred fermenters with strong aeration, because the producing organisms are aerobic. The fermentation has two distinct phases. In the first, the **trophophase**, the organism grows rapidly and uses the easily available nutrients, while little antibiotic is formed. In the second, the **idiophase**, growth slows because a nutrient becomes limiting, and antibiotic production begins. To prolong this production phase, nutrients such as sugar are added slowly (fed-batch), and precursors are supplied. After fermentation, the antibiotic is recovered by filtration, solvent extraction or ion exchange, and purified and crystallized. Every batch is tested for potency, sterility and toxicity.

Penicillin is the best example. It is produced by high-yielding mutants of *Penicillium chrysogenum*. The medium contains corn steep liquor as the main nitrogen source and lactose as the carbon source. Lactose is used because it is broken down slowly, so the fungus does not experience the repression of penicillin synthesis that occurs with abundant glucose; in modern processes glucose is fed slowly instead. The side chain of penicillin G comes from phenylacetic acid, which is added as a precursor. Because phenylacetic acid is toxic, it is fed gradually. The fermentation is carried out at about 25–27°C and pH 6.5 with vigorous aeration for 5–7 days. During the first day or two the fungus grows; then, as growth slows, penicillin accumulates in the broth. The penicillin molecule itself is formed from three amino acids (α-aminoadipic acid, cysteine and valine), which are joined into a tripeptide, cyclized to isopenicillin N, and then given the phenylacetyl side chain.

Penicillin is recovered by filtering off the mycelium, cooling the filtrate and making it acidic (pH 2–2.5). At this pH penicillin is in its un-ionized form and can be extracted into an organic solvent such as butyl acetate. It is then extracted back into a slightly alkaline aqueous buffer, purified with charcoal and crystallized as the potassium or sodium salt. The extraction must be done quickly and in the cold because penicillin is unstable at low pH. Much of the penicillin produced today is converted, using the enzyme penicillin acylase, into 6-aminopenicillanic acid, to which new side chains are attached chemically to make semi-synthetic penicillins such as ampicillin and amoxicillin.

Streptomycin, discovered by Waksman, is produced by *Streptomyces griseus* in a medium containing glucose and soybean meal. It is an aminoglycoside that acts on the bacterial ribosome and was the first effective drug against tuberculosis. The fermentation shows three phases: rapid growth with release of ammonia, then streptomycin production, and finally lysis of the mycelium. Streptomycin is recovered by adsorption on ion-exchange resins or charcoal. **Tetracyclines** are broad-spectrum antibiotics produced by *Streptomyces aureofaciens* and *S. rimosus*; a key point is that chloride must be kept low (or chlorination blocked) to obtain tetracycline rather than chlortetracycline. **Griseofulvin**, produced by *Penicillium griseofulvum*, is an antifungal antibiotic used to treat ringworm; since its molecule contains chlorine, chloride must be added to the medium.

8.1 Introduction

Antibiotics are low-molecular-weight **secondary metabolites** produced by microorganisms that, at low concentrations, inhibit the growth of or kill other microorganisms (Waksman). About 70% of known antibiotics come from **actinomycetes** (mainly *Streptomyces*), about 20% from fungi and 10% from other bacteria (*Bacillus*).

Basis	Classes and examples
Producer	Fungi: penicillin, cephalosporin, griseofulvin; Actinomycetes: streptomycin, tetracycline, erythromycin, chloramphenicol; Bacteria: bacitracin, polymyxin
Spectrum	Narrow (penicillin G: Gram-positive), broad (tetracycline, chloramphenicol), antifungal (griseofulvin, nystatin)

Basis	Classes and examples
Mode of action	Cell wall synthesis (penicillin), protein synthesis (streptomycin, tetracycline), nucleic acid synthesis (rifampicin), cell membrane (polymyxin), mitosis in fungi (griseofulvin)
Chemical structure	β-lactams (penicillins, cephalosporins), aminoglycosides (streptomycin), tetracyclines, macrolides (erythromycin), polyenes (nystatin)

8.2 General steps in antibiotic production

1 Strain selection and preservation (high-yield mutant).

2 Inoculum development: spores → shake flask → seed tanks.

3 Production in a sterile, aerated, agitated fermenter, usually **fed-batch**, with trophophase (growth) followed by idiophase (production).

4 Recovery: separation of mycelium, extraction (solvent or ion exchange), purification, crystallization.

5 Quality control: potency (bioassay), sterility, pyrogen and toxicity tests; formulation.

8.3 Penicillin

Introduction

- β-lactam antibiotic discovered by **Fleming (1928)**. Basic nucleus is **6-aminopenicillanic acid (6-APA)**: a thiazolidine ring fused with a **β-lactam ring**, with a variable side chain (R-CO-).
- Natural penicillins depend on the precursor in the medium: **penicillin G** (benzylpenicillin, side chain from **phenylacetic acid**) and **penicillin V** (phenoxymethylpenicillin, from **phenoxyacetic acid**, acid-stable, oral).
- Action: inhibits **peptidoglycan cross-linking** (binds transpeptidases / penicillin-binding proteins) in growing bacteria; mainly active on Gram-positive bacteria.

Biosynthesis (short)

L-α-aminoadipic acid + L-cysteine + L-valine → (ACV synthetase) → **δ-(L-α-aminoadipyl)-L-cysteinyl-D-valine (ACV tripeptide)** → (isopenicillin N synthase) → **isopenicillin N** → (acyltransferase exchanges side chain with **phenylacetyl-CoA**) → **penicillin G**.

Production process

Stage	Details
Organism	High-yielding mutants of *Penicillium chrysogenum* (descendants of NRRL 1951 / Wis Q-176)
Inoculum	Spores from lyophilized stock → spore suspension on rice or bran → shake flasks → 1–2 seed fermenters; about 5–10% inoculum, **pellet** growth preferred
Production medium	**Corn steep liquor** (N, growth factors, some phenylacetic acid), **lactose** (slowly hydrolysed, so avoids glucose catabolite repression) or glucose / sucrose fed slowly, ammonium salts, **$CaCO_3$** (buffer), KH_2PO_4, $MgSO_4$, **precursor phenylacetic acid** (fed gradually, as it is toxic above about 0.1%), vegetable oil / antifoam
Fermenter	Stirred tank, 40,000–200,000 L, stainless steel; **fed-batch**; sterile air
Conditions	Temperature **25–27°C**; pH **6.5** (kept 6.4–6.8); high aeration (0.5–1.0 vvm) and agitation; 5–7 days (up to 8)
Phase 1: trophophase	First 1–2 days: rapid mycelial growth, lactose/glucose used, pH first falls then rises as NH_3 is released; little penicillin
Phase 2: idiophase	Growth slows; **penicillin formed**; continuous feeding of sugar, ammonia and **precursor**; pH kept near 6.5 with acid/alkali
Phase 3	Nutrients exhausted, autolysis of mycelium, pH rises, penicillin production stops; harvest before this

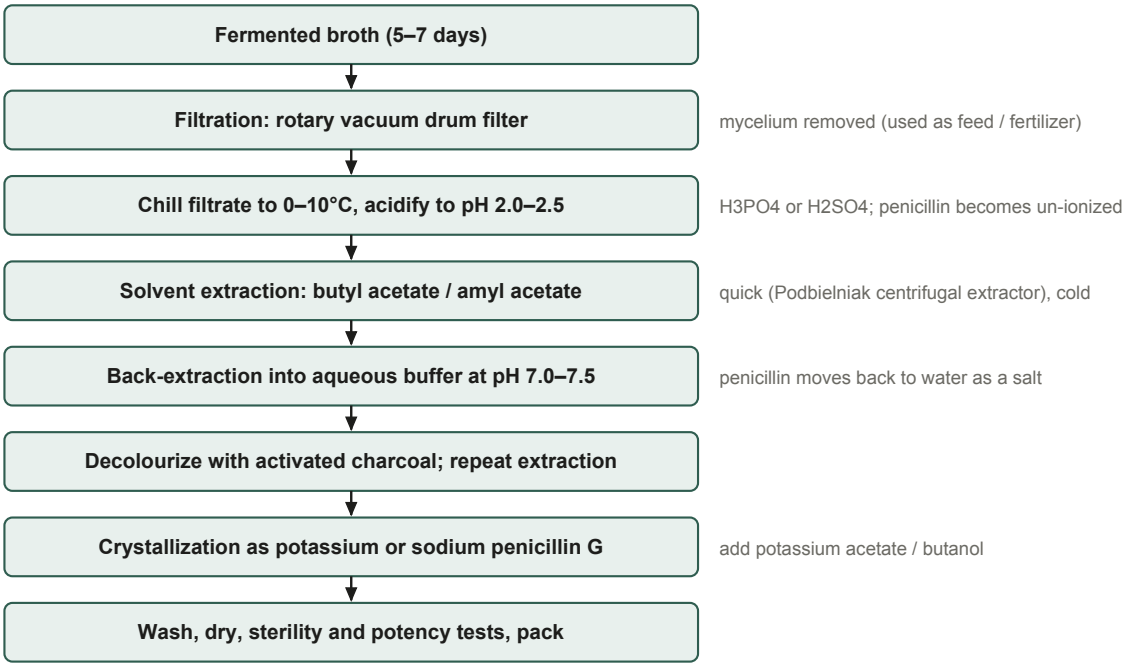


Fig. 7. Recovery (downstream processing) of penicillin

Semi-synthetic penicillins

- Penicillin G is split by **penicillin acylase** (immobilized *E. coli* enzyme) to give **6-APA** and phenylacetic acid.
- New side chains are joined chemically to 6-APA to make **ampicillin, amoxicillin, methicillin, cloxacillin**: broader spectrum, acid stability, or resistance to β-lactamase.

8.4 Streptomycin

- **Discovery:** Waksman, Schatz and Bugie, **1944**, from *Streptomyces griseus*.
- **Structure: aminoglycoside** made of three units: **streptidine** (from glucose via myo-inositol), **streptose** and **N-methyl-L-glucosamine**.
- **Action:** binds the **30S ribosomal subunit**, causes misreading of mRNA; active against Gram-negative bacteria and *Mycobacterium tuberculosis*.
- **Medium:** glucose, **soybean meal** (N source), NaCl, (CSL, distillers' solubles); a strain of *S. griseus* as spores → vegetative inoculum.
- **Conditions:** aerobic submerged, stirred tank, **about 28°C**, pH 7.6–8.0, high aeration, **about 7–10 days**.

Phase	What happens
Phase 1 (about 1 day)	Rapid growth of mycelium; proteolysis of soybean meal releases **NH_3**, so pH rises to about 8.0; glucose used slowly; little streptomycin
Phase 2 (days 2–6)	Little growth; **streptomycin accumulates**; glucose and NH_3 used; pH stays 7.6–8.0
Phase 3	Sugar exhausted; mycelium lyses; production stops; pH rises. Harvest at end of phase 2

- **Recovery:** acidify and filter off mycelium → adsorb streptomycin on **activated charcoal** or a **cation-exchange resin** → elute with dilute acid → concentrate → precipitate / crystallize as **streptomycin sulphate** (or calcium chloride complex) → dry.
- **By-product:** vitamin B_{12} can be recovered from the mycelium.
- **Uses:** tuberculosis (with other drugs), plague, tularaemia; in agriculture against plant bacterial diseases.

8.5 Tetracyclines

- **Chlortetracycline** (aureomycin, 1948, Duggar) from *Streptomyces aureofaciens*; **oxytetracycline** (terramycin) from *S. rimosus*; **tetracycline** from mutants of *S. aureofaciens* (or by removing Cl from chlortetracycline).
- **Structure:** four fused rings (naphthacene); a **polyketide** made from acetate/malonate units.

- **Action:** bind the **30S subunit** and block attachment of aminoacyl-tRNA; **broad spectrum** (Gram-positive, Gram-negative, rickettsiae, chlamydiae, mycoplasma).
- **Medium:** starch or sucrose, CSL, soybean meal, $CaCO_3$, $(NH_4)_2SO_4$; **phosphate kept low** (it represses production).
- **Key control:** to get **tetracycline** instead of chlortetracycline, **chloride is kept low** in the medium or a **chlorination inhibitor** (bromide, mercaptobenzothiazole) is added, or a chlorination-deficient mutant is used.
- **Conditions:** aerobic submerged, about 28°C, pH about 6.0–6.8, 4–6 days, fed-batch.
- **Recovery:** acidify broth (tetracycline binds Ca/Mg, so oxalic acid is added to release it), filter, extract with butanol or precipitate as a salt, purify and crystallize (as hydrochloride).

8.6 Griseofulvin

- **Antifungal** antibiotic from *Penicillium griseofulvum* (also *P. patulum*, *P. urticae*); discovered 1939 (Oxford, Raistrick).
- **Structure:** chlorine-containing **polyketide** (spiro benzofuran); a **chlorine atom** is part of the molecule.
- **Action:** binds **tubulin** and disrupts the mitotic spindle, stopping fungal cell division; deposited in keratin of skin, hair and nails.
- **Production:** aerobic submerged fed-batch fermentation, about 25°C, pH about 6–7, 7–10 days; medium with CSL, lactose or glucose, and **chloride (KCl)** added because it is incorporated into griseofulvin.
- **Recovery:** product is mostly in the mycelium: filter, extract mycelium with organic solvent (butyl acetate / acetone), concentrate and crystallize.
- **Uses:** oral treatment of **ringworm** (dermatophytes: *Trichophyton*, *Microsporum*, *Epidermophyton*) of skin, hair and nails; not active against bacteria or yeasts.

Antibiotic	Producer	Class	Target	Key production point
Penicillin G	*P. chrysogenum*	β-lactam	Cell wall	Phenylacetic acid precursor, lactose, fed-batch, pH 2–2.5 solvent extraction
Streptomycin	*S. griseus*	Aminoglycoside	30S ribosome	Soybean meal, 3 phases, cation exchange recovery
Tetracycline	*S. aureofaciens*, *S. rimosus*	Tetracycline (polyketide)	30S ribosome	Low chloride and low phosphate
Griseofulvin	*P. griseofulvum*	Polyketide (Cl)	Fungal tubulin	Chloride added; extracted from mycelium

Exam point: Penicillin is the most asked long question: organism → structure → inoculum → medium with precursor → fed-batch conditions → phases → recovery flow chart (Fig. 7) → semi-synthetic penicillins. Short notes on the other three from the summary table.

9. VITAMIN PRODUCTION

Detailed explanation

Vitamins are organic compounds that the human body needs in very small amounts but cannot make for itself, so they must be obtained from food or supplements. Many vitamins are now made by chemical synthesis, but some are produced more efficiently or only by microorganisms. The two most important examples, riboflavin (vitamin B_2) and cyanocobalamin (vitamin B_{12}), are included in the syllabus.

Riboflavin is a yellow vitamin that forms part of the coenzymes FMN and FAD, which take part in many oxidation–reduction reactions of the cell. Some fungi, especially *Eremothecium ashbyii* and *Ashbya gossypii*, naturally overproduce riboflavin in such large amounts that it crystallizes inside and around the cells. These fungi were the basis of the first industrial processes. The fungus is grown in a well-aerated submerged fermentation at about 26–28°C, using sugars or plant oils as the carbon source together with nitrogen-rich materials such as corn steep liquor, peptone or gelatin. Riboflavin is formed mainly after the growth phase. At the end of the fermentation the broth is heated to release riboflavin from the cells, and the crystals are collected and purified. Today, genetically engineered strains of *Bacillus subtilis* and improved *Ashbya* strains produce riboflavin very efficiently, and microbial production has largely replaced chemical synthesis. Most riboflavin is used in animal feed, and some as a food colour and in vitamin tablets.

Vitamin B_{12} is the most complex vitamin. Its molecule contains a corrin ring with a cobalt atom at the centre and a lower group called dimethylbenzimidazole. It is needed for the formation of red blood cells and for the nervous system, and its deficiency causes pernicious anaemia. Remarkably, vitamin B_{12} is made only by certain bacteria and archaea; plants and animals cannot make it. Therefore all commercial vitamin B_{12} is produced by fermentation. The main organisms are *Propionibacterium freudenreichii* (and *P. shermanii*) and *Pseudomonas denitrificans*.

In the *Propionibacterium* process, the medium contains glucose, corn steep liquor and a cobalt salt. The fermentation is carried out in two stages. In the first stage, lasting a few days, the culture is kept **anaerobic**; the bacteria grow and form the corrin part of the molecule. In the second stage the culture is **aerated**; this allows the formation of dimethylbenzimidazole (which may also be added), which is attached to complete the vitamin. In the *Pseudomonas* process the fermentation is aerobic throughout, and beet molasses is used because it contains betaine, which greatly increases the yield. Since vitamin B_{12} remains inside the cells, the cells are harvested and heated to release it, and cyanide is added to convert all the different forms into stable cyanocobalamin, which is then purified and crystallized. Vitamin B_{12} is used to treat pernicious anaemia and as a supplement in human diets and animal feeds.

Vitamins are organic compounds needed in small amounts that the body cannot make. Some are made chemically; **riboflavin** and **vitamin B_{12}** are made mainly by **microbial fermentation** (B_{12} is made **only** by microbes).

9.1 Riboflavin (vitamin B_2)

- **Structure and role:** yellow, water-soluble vitamin (isoalloxazine ring + ribitol); forms the coenzymes **FMN** and **FAD** used in redox reactions. Deficiency: ariboflavinosis (cheilosis, glossitis).
- **Producers:** the ascomycete fungi ***Eremothecium ashbyii*** and ***Ashbya gossypii*** (natural overproducers, first used in the 1940s); the yeast *Candida famata* (flareri); and today mostly **genetically engineered *Bacillus subtilis*** and improved *A. gossypii*. Earlier, *Clostridium acetobutylicum* gave riboflavin as a by-product of acetone–butanol fermentation.
- **Biosynthesis:** from **GTP** and **ribulose-5-phosphate**, via 6,7-dimethyl-8-ribityllumazine; glycine and purine precursors increase yield.

Stage	Details (*Ashbya gossypii* process)
Inoculum	Stock → shake flasks → seed tank (about 5%)
Medium	Glucose or sucrose, or **plant oils** (soybean oil); **corn steep liquor**, peptone, collagen or gelatin, yeast extract; glycine (precursor); inositol; no iron limitation needed (unlike *Candida*, which needs **low iron**)
Conditions	Aerobic submerged, stirred tank, about **26–28°C**, pH 6–7, high aeration, about **5–7 days** (fed-batch)

Stage	Details (*Ashbya gossypii* process)
Phases	Growth phase first; riboflavin made mostly after growth slows (**stationary phase**); much of it **crystallizes** in the cells and broth as yellow crystals
Recovery	Heat broth (about 1 h at 120°C) to release riboflavin from cells → centrifuge → collect crystals / precipitate → purify by recrystallization → dry. For feed grade, whole broth is dried
Yield	Modern processes give over 10–20 g/L
Uses	Animal feed supplement (most), human vitamin tablets, **yellow food colour** (E101)

9.2 Cyanocobalamin (vitamin B_{12})

- **Structure:** the largest and most complex vitamin: a **corrin ring** with a central **cobalt** atom, a lower ligand **5,6-dimethylbenzimidazole (DMB)** and an upper ligand (cyano-, methyl-, adenosyl- or hydroxo-). Commercial form is **cyanocobalamin**.
- **Role:** coenzyme in methionine synthase and methylmalonyl-CoA mutase; needed for red blood cell formation and nerves. Deficiency: **pernicious anaemia**.
- Made **only by bacteria and archaea**, not by plants or animals, so it is produced entirely by fermentation. First obtained as a by-product of streptomycin fermentation.
- **Producers:** ***Propionibacterium freudenreichii*** (and *P. shermanii*) and ***Pseudomonas denitrificans*** (main industrial organisms, highly improved strains); also *Bacillus megaterium*, *Streptomyces*.

Point	*Propionibacterium freudenreichii* process	*Pseudomonas denitrificans* process
Medium	Glucose, **corn steep liquor**, **cobalt** salt ($CoCl_2$), **DMB** added later	**Sugar beet molasses** (contains **betaine**, which strongly increases yield), CSL, cobalt, DMB
Conditions	**Two-stage**, about 30°C, pH about 7 (neutralized with NH_4OH)	Aerobic throughout, about 30°C, pH 6–7
Stage 1	**Anaerobic** for 2–4 days: growth and formation of the corrin precursor (cobinamide)	Single aerobic stage, fed-batch
Stage 2	**Aerobic** for 3–4 days: DMB is made (or added) and linked, forming complete B_{12}	—
Duration	About 6–7 days	About 5–7 days

- **Recovery:** B_{12} is **intracellular** → cells harvested by centrifugation → heated (80–120°C) at pH 6.5–8.5 to release the vitamin → **potassium cyanide** added to convert all cobalamins to stable **cyanocobalamin** → adsorption/solvent extraction and chromatography → crystallization. Feed grade: concentrated broth dried.
- **Uses:** treatment of **pernicious anaemia** and B_{12} deficiency, vitamin supplements (vegetarians), animal feed.

9.3 Other vitamins (extra marks)

- **Vitamin C (ascorbic acid)**: **Reichstein process**: D-glucose → (chemical hydrogenation) D-sorbitol → (***Gluconobacter oxydans*** oxidation) L-sorbose → chemical steps → ascorbic acid. The newer two-step fermentation uses *Gluconobacter* and *Ketogulonicigenium vulgare* to make 2-keto-L-gulonic acid.
- **β-Carotene (provitamin A)**: fungus *Blakeslea trispora* (mixed + and − mating strains), microalga *Dunaliella salina*.
- **Vitamin D**: ergosterol from yeast irradiated with UV gives vitamin D_2.

Exam point: Riboflavin: organisms (Ashbya, Eremothecium, B. subtilis) → medium → conditions → crystallization → recovery → uses. B_{12}: structure with cobalt and DMB → organisms → two-stage anaerobic–aerobic Propionibacterium process → betaine in Pseudomonas → cyanide in recovery → pernicious anaemia.

10. QUICK REVISION AND LIKELY QUESTIONS

Remember	Answer
Term "biotechnology"	Karl Ereky, 1919
Father of industrial microbiology	Louis Pasteur (fermentation by microbes, 1857)
Cell-free fermentation	Eduard Buchner, 1897 (zymase)
Acetone–butanol (WWI)	Chaim Weizmann, *Clostridium acetobutylicum*
Citric acid (industrial, 1923)	Pfizer, *Aspergillus niger* surface culture (Currie)
Penicillin	Fleming 1928; Florey and Chain; Peoria deep-tank, CSL, NRRL 1951
Streptomycin	Waksman, Schatz, Bugie 1944, *S. griseus*
First rDNA / first cloned human protein / first rDNA drug	Cohen–Boyer 1973 / somatostatin 1977 / Humulin 1982
Six components of fermentation	Media, sterilization, inoculum, production, downstream, effluent
Trophophase / idiophase	Growth phase (primary metabolites) / production phase (secondary metabolites)
Baffles	Usually 4, width 1/10 of diameter; prevent vortex
Airlift zones	Riser (gassed) and downcomer; draft tube; internal or external loop
Fluidized vs packed bed	Particles suspended by upward flow vs fixed bed
Chemostat	D = F/V; steady state $\mu = D$; washout if $D > \mu_{max}$
Monod equation	$\mu = \mu_{max} \cdot S / (K_s + S)$
Fed-batch avoids	Substrate inhibition, catabolite repression, Crabtree effect
SSF	Little free water, fungi, wheat bran, koji; a_w important
Enzyme unit	1 µmol substrate converted per minute; SI unit katal
First immobilized enzyme / first industrial	Invertase on charcoal 1916 / aminoacylase 1969 (Chibata)
Entrapment gel / cross-linker	Calcium alginate / glutaraldehyde
Largest immobilized enzyme use	Glucose isomerase → high fructose corn syrup
Crowded plate / auxanography	Antibiotic producers / growth-factor producers
Most potent chemical mutagen	NTG (GC→AT transitions)
Protoplast fusion agent	Polyethylene glycol (PEG)
Penicillin precursor / nucleus	Phenylacetic acid (Pen G) / 6-APA
Penicillin extraction	pH 2–2.5, butyl or amyl acetate, back-extract at pH 7.5
Streptomycin parts	Streptidine + streptose + N-methyl-L-glucosamine
Tetracycline control	Low chloride (or bromide) and low phosphate
Griseofulvin	*P. griseofulvum*; contains Cl; antifungal (tubulin); ringworm
Riboflavin producers	*Ashbya gossypii*, *Eremothecium ashbyii*, *B. subtilis*
B_{12}	*Propionibacterium* (anaerobic → aerobic), *Pseudomonas denitrificans* (betaine); cobalt + DMB; cyanide in recovery

Likely long questions

1 Define biotechnology. Trace the history of microbial biotechnology with important milestones.

2 What are the components of a fermentation process? Differentiate primary and secondary metabolites.

3 Describe the design and parts of a stirred tank fermenter with a labelled diagram.

4 Describe airlift, bubble column, fluidized bed, packed bed, hollow fibre and photobioreactors.

5 Differentiate batch, fed-batch and continuous fermentation. Explain the chemostat.

6 Compare submerged and solid state fermentation with examples.

7 Describe the industrial production of microbial enzymes and their therapeutic, analytical and industrial applications.

8 What is enzyme immobilization? Describe the methods with advantages and disadvantages and give applications.

9 Describe primary and secondary screening. Explain the methods of strain improvement.

10 Describe the industrial production and recovery of penicillin.

11 Write notes on production of streptomycin, tetracycline and griseofulvin.

12 Describe the microbial production of riboflavin and vitamin B_{12}.

Likely short notes (2–5 marks)

- Karl Ereky · Weizmann process · Peoria contribution · Buchner's experiment
- Baffles · sparger · Rushton turbine · $k_L a$ · washout · turbidostat
- Koji · water activity in SSF · enzyme unit · catabolite repression
- Calcium alginate beads · CLEA · glucose isomerase
- Crowded plate · auxanography · NTG · protoplast fusion · replica plating
- 6-APA · semi-synthetic penicillins · streptidine · chloride in tetracycline and griseofulvin
- Betaine and B_{12} · DMB · Ashbya gossypii

References: Stanbury, Whitaker and Hall, Principles of Fermentation Technology; Casida, Industrial Microbiology; Patel, Industrial Microbiology; Prescott and Dunn's Industrial Microbiology; Crueger and Crueger, Biotechnology: A Textbook of Industrial Microbiology; class notes "History of Biotechnology" and "Bioreactor Design, Unit 5" (Dr. Annam Renita A.).